Cartier–Miller evaluation of genus-one coefficient sums

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Abstract

Let $p \geq 7$ be prime and let $f \in \mathbb{F}_p[w]$ be squarefree of degree three or four with $f(0) = 1$. We express every admissible base-field value of the truncation $T_f = f^{(p-1)/2} \bmod w^p$ through the elliptic trace and a normalized logarithmic derivative. The identity includes trace ± 1, with the residue-divisor normalization proved directly in characteristic p for both signs. A differentiated Miller recurrence gives $O(\log p)$ field operations per query after trace preparation, and Schoofs algorithm gives deterministic polynomial-time preparation. A quadratic algebra avoids square-root and nonresidue searches. A uniform curve-order Miller identity gives a second evaluator for nonbranch rationally weighted quadratic-character sums, valid also for $p = 5$. Supplied simple-pole weights and equivalent elliptic-point sums have explicit common-domain corrections and bit bounds. The construction uses classical Cartier, logarithmic-transfer and local-coefficient machinery; publication priority for the complete scalar formula is not established. Matched-output experiments compare sparse, dense and cached workloads. The original quarter-point weighted central-binomial sum is a worked application.

Introduction

We study exact evaluation modulo a prime of coefficient sums attached to cubic and quartic polynomial powers. Such a power has a number of coefficients proportional to the prime, although the associated curve has genus one. Our aim is to evaluate a specified weighted sum of these coefficients through elliptic-curve arithmetic, with the cost of preparing the curve separated from the cost of each evaluation.

Let $p \geq 7$ be prime, put $h = (p-1)/2$, and let $f \in \mathbb{F}_p[w]$ be squarefree of degree three or four, with $f(0) = 1$. Write

$$f(w)^h = \sum_m c_m w^m, \qquad T_f(w) = \sum_{m=0}^{p-1} c_m w^m, \qquad H = c_{p-1}. \qquad (1)$$

For $\lambda \neq 0$ in $\mathbb{F}_p$, consider the weighted coefficient expression

$$S_f(\lambda) = [w^{p-1}] \frac{f(w)^h}{1-\lambda w} = \sum_{m=0}^{p-1} c_m \lambda^{p-1-m} = T_f(1/\lambda). \qquad (2)$$

The last equality uses $\lambda^{p-1} = 1$ and is a base-field identity. At $\lambda = 0$, the coefficient expression gives $S_f(0) = H$; evaluation $T_f(0) = 1$ is separately immediate. Thus varying the rational weight amounts to evaluating one fixed truncation at different base-field arguments. We call an argument admissible when $\lambda = 0$, or when $\lambda \neq 0$ and $f(\lambda^{-1}) \neq 0$. The latter condition excludes the branch points of the double cover $v^2 = f(w)$. A polynomial with any nonzero constant term reduces to this normalization by dividing by that term and restoring its hth power afterwards, as detailed in Corollary 4. Euler's criterion also gives $T_f(\mu) = \chi - \sum_{m \geq p} c_m \mu^m$, where $\mu = \lambda^{-1}$ and $\chi = (f(\mu)/p)$ at a nonzero admissible argument.

The classical Hasse–Witt congruence identifies H with the elliptic trace modulo p. We extend this coefficient selection to the family of differentials $dw/\big((1-\lambda w)v\big)$, whose nonzero admissible members have two simple poles. Cancelling their residues with a logarithmic differential leaves a regular differential. In genus one, that remainder is a scalar multiple of the fixed invariant differential, so evaluation at one point determines the correction. Theorem 1 makes this observation into an explicit coefficient-sum identity, including the exceptional cases in which the relevant curve or quadratic twist has exactly p rational points.

Truncated binomial and hypergeometric congruences motivate the evaluation problem. Direct accumulation of a suitable coefficient recurrence at a cutoff proportional to p takes $\Theta(p)$ field operations, while polynomial-block methods provide approximately square-root dependence, up to logarithmic factors, in suitable recurrence problems (Bostan et al., 2007; Harvey, 2007, Theorem 10). The selected-coefficient adapter (W10)–(W11) gives such a trace-free comparator for our exact character-sum output, including branch poles. Our logarithmic query bound assumes a prepared exact trace; a first query on a new curve includes point counting. Repeated queries on one curve reuse that preparation. Tables across many primes form another workload, for which accumulating remainder forests can share recurrence products (Harvey & Sutherland, 2014, 2016).

The main theorem fixes the exponent at $(p-1)/2$, the coefficient cutoff at $p-1$, and the output modulo p. It varies the admissible base-field argument. The binomial applications below require both a prefix and a boundary coefficient; identifying a fast prefix does not by itself give a fast complete weighted sum. The implementation and benchmark sections distinguish these outputs and workloads. The character-sum application also admits arbitrary constant term and a pole at zero: Lemma 6 and Theorem 8 give a rational base-field chain at the order of a normalized twist, and Theorem 9 accounts for every excluded fibre in a supplied simple-pole weight.

Theoretical contributions

The main result is a uniform identity for admissible base-field coefficient sums. Let E be the smooth projective curve $v^2 = f(w)$ with origin $O = P_+ = (0,1)$, invariant differential $\omega = dw/v$, and exact trace $\tau = p + 1 - \#E(\mathbb{F}_p)$. For a nonzero admissible argument define

$$\begin{gathered} d = f(1/\lambda), \quad v_0^2 = d, \quad \chi = (d/p), \\ Q_\pm = (1/\lambda, \pm v_0), \quad D = (Q_+) - (Q_-), \quad r = \frac{1}{\lambda v_0}. \end{gathered} \tag{3}$$

For an integer M annihilating the class of D, fix a rational function F_D with $\operatorname{div}(F_D) = MD$, and set $\Psi_M(D) = \big((dF_D/F_D)/\omega\big)(O)$. This value is independent of the scalar multiplying F_D. For $p \nmid M$ it retains the specified divisor representative; for $M = p$ it depends only on its divisor class.

Theorem 1 (Uniform base-field evaluation). Under these hypotheses, $H \equiv \tau \pmod p$, and every nonzero admissible argument satisfies

$$S_f(\lambda) = H - \chi r \Psi_{M_\chi}(D), \qquad M_\chi = p + 1 - \chi\tau. \tag{4}$$

The product $r\Psi_{M_\chi}(D)$ belongs to $\mathbb{F}_p$ and is unchanged by replacing v_0 by $-v_0$. The formula includes $\tau = \chi$, when $M_\chi = p$, and requires no division by that annihilator. Corollary 4 states the evaluation and preparation costs.

The proof begins with coefficient selection by the Cartier operator. Proposition 2 gives the regular residue identity. Lemma 3 supplies the additional normalization when $\tau = \chi$, identifying the Cartier defect with the established logarithmic transfer on geometric p-torsion. The proof fixes the differential and translation conventions directly in characteristic p, including the negative Frobenius eigenspace at trace -1. The mathematical focus is the coefficient-sum formula with this specified scalar and residue divisor. The underlying additive map, local-coefficient cocycle and differentiated Miller machinery are established constructions, discussed in the next section. Lemma 6 gives an additional all-trace derivation with the curve order as scalar. It explains the annihilator and yields the nonbranch character-sum adapter without a trace case split; its publication priority remains unresolved.

The identity is constructive. A differentiated Miller chain computes its logarithmic scalar in $O(\log p)$ field operations after trace preparation. A quadratic $\mathbb{F}_p$-algebra implements the conjugate calculations without a preliminary square-root or quadratic-nonresidue search; in the split case, a zero divisor supplies the splitting root. Corollary 4 combines this query algorithm with Schoof's deterministic point count to prove polynomial-time evaluation in $\log p$, and states the separate preparation and query bit costs. The specialized preparation used in the computational examples is distinguished there from that worst-case guarantee.

The same coefficient identity evaluates a rationally weighted quadratic character sum.

Corollary 1.1 (Weighted quadratic character sum). Let $\mu \in \mathbb{F}_p^{\times}$ with $f(\mu) \neq 0$, put $\lambda = \mu^{-1}$ and use the data of Theorem 1. Define $\varepsilon_f(w) = \left(\frac{f(w)}{p}\right)$, including the value zero at a root of f, and put $\chi = \varepsilon_f(\mu)$. Then

$$\sum_{\substack{w \in \mathbb{F}_p \\ w \neq \mu}} \frac{\varepsilon_f(w)}{w - \mu} = \frac{H - \chi}{\mu} + \frac{\chi f'(\mu)}{2f(\mu)} - \frac{\chi}{v_0} \Psi_{M_\chi}(D). \qquad (5)$$

This character sum has the same prepared-query cost. The independent point-sum identity (51), proved in the section on scope, gives the corollary on substituting (4) and supplies a direct verification route with no Miller functions. The later application section proves the arbitrary-constant-term and zero-pole extension, constant-plus-simple-pole weights, their equivalence with elliptic-point sums, and the complete excluded-domain and infinity corrections. Corollary 10 separates field-operation and bit costs from trace acquisition; the prepared-query bound is not a logarithmic bit bound.

The binomial applications show how the geometric identity interacts with additional arithmetic. At the original quarter-point cutoff, classical complex-multiplication data supply the trace and boundary, while Theorem 1 supplies the prefix. At the rounded-third cutoff, the prefix has known zero trace, but the boundary remains equivalent to a modular factorial through the explicit inverse formulas (43). Proposition 5 locates that boundary in full de Rham cohomology in a specified basis, even though Cartier on regular differentials is zero. The complete evaluator retains the boundary-acquisition cost; a separate many-prime recurrence acquires prefix and boundary jointly. Their formulas, proofs and matched computational comparisons are given in the application sections and appendix.

Relation to prior work

The geometric ingredients of the evaluation are classical. The Artin–Schreier construction underlying the high polynomial block is traced to Deuring (1941) in the accounts of Voloch (1990, Lemma 1.1; 1997, Section 2). Cartier (1957) established the differential operator used here, and Serre (1958, Propositions 8 and 10) describes its properties and the correspondence between geometric p-torsion divisor classes and Cartier-fixed holomorphic logarithmic differentials. These results provide the coefficient-selection and logarithmic-transfer framework for the present paper.

Cassels (1971, Sections 1 and 5–6) develops the high-block Artin–Schreier equation and its analogy with elliptic quasi-periods. Gunji (1976, equations (2.4), (2.9), and Lemma 2.11) studies functions with divisor $p\big((P) - (O)\big)$, their logarithmic differentials and Cartier eigenvalues on ordinary elliptic curves. His setting is not restricted to trace-one curves. His differential is dx/y, whereas our short model uses $\omega_0 = dx/(2y)$; this factor must be retained when comparing scalars. The rounded-third de Rham interpretation also sits within the classical relationship between de Rham cohomology, Frobenius and Dieudonné theory developed by Oda (1969).

Trace acquisition is a separate classical algorithmic task. Schoof (1985) gives a deterministic polynomial-time point-counting algorithm; Schoof (1995) surveys the method and subsequent developments. This supplies the preparation guarantee used in Corollary 4. Our $O(\log p)$ field-operation bound concerns a query after an exact trace has been supplied or computed. It does not replace point counting or describe the bit cost of the complete calculation. The specialized CM preparation and the practical trace providers used in the experiments have their own costs and hypotheses.

Miller's foundational contribution is the short arithmetic representation of rational functions with prescribed divisors. His unpublished manuscript dated 6 May 1986 constructs these functions as straight-line programs through divisor addition, avoiding their full polynomial expansion (Miller, 1986, Section 2). The later published exposition (Miller, 2004, Section 4.1, especially equations (7)–(9)) gives the divisor, addition and composition identities for the functions $f_{m,P}$, together with addition-chain evaluation in $O(\log m)$ elliptic additions. These are the functions and identities used in our differentiated recurrence and in Lemma 6. The 2004 publication is therefore cited as a later exposition of machinery already available before Semaev's 1998 and Rück's 1999 papers. Our computation extracts a local coefficient of $d\log f_{m,P}$ from that circuit; the short function construction itself is not a contribution of this paper.

Voloch (1990, Lemma 1.1 and Proposition 1.3) identifies the Artin–Schreier cover with Verschiebung and obtains an explicit p-descent homomorphism. Voloch (1997, Propositions 1–3) develops the characteristic-p zeta function, its addition law and its relation to logarithmic differentials, while correcting scale and sign conventions in the earlier account. On a trace-one short curve, the descent function is represented by the product of the ordinate and the high block of the cubic power. This supplies a close structural antecedent for (24)–(25). The differential scale, uniformizer and actual divisor representative must still be specified when comparing that polynomial with a Miller coefficient.

Semaev (1998, Lemmas 1–3) gives the directly relevant computational specialization: an additive logarithmic-derivative embedding of an order-p subgroup, evaluated by differentiating the function recurrence in $O(\log p)$ field operations. Rück (1999, p. 806) extends the transfer to divisor class groups and explicitly constructs the local-coefficient cocycle for divisor addition. His addition-function differential may have a pole at the expansion point; in genus one the needed coefficient is its Laurent constant. Repeated doubling computes the transfer. Thus both differentiated Miller evaluation and its local-coefficient cocycle interpretation have explicit antecedents. Galbraith (2012, Section 26.4.1) gives a later account of the p-torsion differential map and its coefficient computation.

The anomalous-curve algorithms form the corresponding cryptographic context. Satoh and Araki (1998, with the 1999 correction) use elliptic Fermat quotients, and Smart (1999) uses a p-adic lift, to solve discrete logarithms when $\#E(\mathbb{F}_p) = p$. Blake, Seroussi and Smart (1999, Chapter V, Section 3) explain the lifting approach. Trace -1 reduces to trace one on a quadratic twist. These order-p transfers account for the exceptional setting of Theorem 1; they do not make the general-trace Laurent coefficient an additive discrete-logarithm embedding on an arbitrary elliptic curve. Lemma 3 fixes the normalization relative to $\omega = dw/v$ and the specified two-pole residue divisor, including the factor of two associated with $(P) - (-P)$. Lemma 6 gives an all-trace identity using the rational curve order as the Miller scalar.

The coordinate and algebraic implementation also uses established constructions. The cubic/quartic model change follows Connell (1999, Section 1.2), with its differential scale made explicit here. Splitting the quadratic algebra when a zero divisor appears is a degree-two instance of dynamic evaluation, or the D5 principle (Della Dora et al., 1985). These ingredients remove particular root-search requirements; they do not constitute a new general root-finding algorithm.

A separate computational lineage provides the natural trace-free comparison. Bostan, Gaudry and Schost (2007) accelerate polynomial recurrences through block products. Harvey (2007, Theorem 10) gives the interval-product theorem used by our selected-coefficient adapter. For fixed degree, (W10)–(W11) give a $\tilde{O}(\sqrt{p})$-field-operation route to the same single-pole output, including branch poles, with interpolation and denominator preparation charged. The block-product machinery is classical; its conversion to the weighted-sum output is stated and proved here. Its trace-free setup must be distinguished from the trace-supplied query model and from dense-query polynomial methods.

Harvey and Sutherland (2014, 2016) address another workload, computing Hasse–Witt matrices over many primes by accumulating remainder trees and forests. The rounded-third companion specializes that machinery to a triangular recurrence acquiring prefix and boundary jointly. Asif, Fité and Pentland (2022), with an appendix by A. V. Sutherland, use Cartier–Manin data and supplementary splitting information to reconstruct L-polynomials of generic Picard curves outside a density-zero set of good primes. Their genus-three reconstruction provides a methodological comparison with our genus-one coefficient evaluation; it does not supply the residue-divisor identity. Frobenius on third-kind differentials in local height computations (Balakrishnan & Besser, 2012) and Cartier coefficient calculations in Dwork crystals (Beukers & Vlasenko, 2021) are additional related frameworks, with different outputs and hypotheses.

The arithmetic applications require further input beyond a fast prefix. The quarter-point boundary uses classical Gauss/CM congruences, with the normalization recorded by Cosgrave and Dilcher (2010); truncated central-binomial congruences are also studied by Sun (2011). The rounded-third boundary uses Jacobi-sum methods of van Wamelen (2002) and the split construction considered in Tal's unrefereed preprint (2026), with a prime conductor dividing $p - 1$. Our implementation specializes the anchor intervals and verifies reconstruction certificates. It does not inherit the exponent claim for Tal's distinct extension-field construction, or turn a fast prefix into a fast acquisition of the missing boundary coefficient.

An elementary multiplicative analogy is the finite logarithm (Elbaz-Vincent & Gangl, 2002):

$$\ell_{1,p}(X) = \sum_{j=1}^{p-1} \frac{X^j}{j} \equiv \frac{1 - X^p - (1-X)^p}{p} \pmod{p}. \qquad (6)$$

The quotient is formed in $\mathbb{Z}[X]$ before reduction. At a base-field argument, exponentiation modulo p^2 and exact division by p evaluate it with $O(\log p)$ ring operations. This is a comparison with short arithmetic evaluation of a long truncation, without identifying our regular elliptic coefficient with a multiplicative Fermat quotient.

The mathematical focus here is the explicit connection between coefficient sums, normalized logarithmic scalars and rationally weighted character sums, including the cubic/quartic adapters, complete excluded domains, equivalent elliptic-point sums and separately charged field and bit costs. Theorem 1 and Lemmas 3 and 6 specify those connections and their normalizations. We do not claim novelty for Miller's short-chain construction, the p-torsion logarithmic transfer or Rück's local-coefficient cocycle. We have not located the complete general-trace scalar identity with its weighted-sum adapters in the sources examined; this bounded comparison establishes neither historical priority nor prior publication of that combined evaluator.

Notation

Table 1. Principal notation and local conventions.

Symbol	Meaning
f, w, v, O, ω	Normalized cubic or quartic model, coordinates, distinguished origin and fixed differential
T_f, S_f, H, τ	Truncation, coefficient sum, Hasse coefficient and integer trace
$\pi, \sigma, \mathcal{C}$	Point Frobenius, coefficient conjugation and Cartier operator
χ, ε_f	Character at the argument and the quadratic character function
D, M_χ, Ψ_M	Specified two-pole divisor, annihilator and normalized logarithmic derivative
K, Θ_R	Torsion kernel and third-kind differential in Lemma 3
$\mathcal{G}_f, g$	High-part value and a rational function defining a principal change
$a_i, B_L, U_p(L)$	Central-binomial coefficient, strict prefix and weighted sum
m, L, G	Rounded-third indices and $G = m! \bmod p$
ℓ, k_ℓ, N	Prime Jacobi conductor, grid step $(p-1)/\ell$ and denominator-12 anchor index
$\mathcal{T}, \mathcal{S}, \mathcal{D}$	Low and high cubic-power parts and differential operator in Proposition 5
Δ_j, A_j	Cleared factorial denominator and forest recurrence matrix
$N_{\mathrm{qry}}, q_k, e_k$	Number of queries, extension-field size and extension character sign

Cartier evaluation and the regular identity

All differential identities are in characteristic p. Frobenius π on points means the p-power map on coordinates. Coefficient conjugation σ raises constants to their pth powers while leaving coordinate variables fixed. Put

$$\omega = \frac{dw}{v}, \qquad \theta_\lambda = \frac{dw}{(1-\lambda w)v}. \qquad (7)$$

The differential ω is holomorphic and nowhere zero. For nonzero admissible λ, the only poles of θ_λ are simple poles at Q_+ and Q_- with residues $-r$ and r.

Let $\mathcal{C}$ denote absolute Cartier, and let id denote the identity on rational differentials. We use the local convention

$$\mathcal{C}\left(\sum_n b_n z^n \, dz\right) = \sum_j b_{pj+p-1}^{1/p} z^j \, dz. \qquad (8)$$

It satisfies $\mathcal{C}(a^p\eta) = a\mathcal{C}(\eta)$ and $\mathcal{C}(dF/F) = dF/F$ (Cartier, 1957; Serre, 1958, Proposition 8). Since $f(0) = 1$, w is a uniformiser at O. The relation $v^{p-1} = f^h$ gives the coefficient selection directly:

$$\theta_\lambda = v^{-p}\frac{f^h}{1-\lambda w}\,dw, \qquad \mathcal{C}(\omega) = H\omega, \qquad \left(\frac{\mathcal{C}(\theta_\lambda)}{\omega}\right)(O) = S_f(\lambda)^{1/p}. \tag{9}$$

Here $H \in \mathbb{F}_p$, so $H^{1/p} = H$. The trace relation is the genus-one Hasse–Witt identity (Silverman, 2009, Chapter V), and can also be checked by the power sums

$$\sum_{w\in\mathbb{F}_p} w^k = \begin{cases} -1, & k > 0 \text{ and } p-1 \mid k, \\ 0, & \text{otherwise,} \end{cases} \qquad \text{in } \mathbb{F}_p. \tag{10}$$

For a cubic, only c_{p-1} survives in the sum of $f(w)^h$, and the projective curve has one point at infinity. For a quartic, $c_{2p-2} = f_4^h = (f_4/p)$ also survives; the infinity contribution is $1 + (f_4/p)$ and cancels this extra term. In both cases $\tau \equiv H \pmod p$.

Proposition 2 (Regular residue identity). Let $\lambda \neq 0$ be in any perfect extension of $\mathbb{F}_p$, with $f(1/\lambda) \neq 0$. Suppose $M \geq 1$, $p \nmid M$, and fix F_D with $\operatorname{div}(F_D) = MD$ as divisors. Then

$$M\big(S_f(\lambda) - H\big) = r(H\Psi_M(D) - \Psi_M(D)^p). \tag{11}$$

Proof. The residues cancel in $M\theta_\lambda + r\,dF_D/F_D$, so this differential is $c\omega$ for some scalar c. Evaluation at O gives $c = M + r\Psi_M(D)$. Apply Cartier, evaluate at O, and raise to the pth power to obtain

$$MS_f(\lambda) + r\Psi_M(D)^p = \big(M + r\Psi_M(D)\big)H. \tag{12}$$

This is (11). For the fixed divisor D, replacing F_D by F_D^k gives $\Psi_{kM}(D) = k\Psi_M(D)$ in characteristic p.

For a base-field argument, $\pi(Q_+) = Q_\chi$, so $\pi(D) = \chi D$ as divisors. Frobenius's characteristic equation annihilates its divisor class by $M_\chi = p + 1 - \chi\tau$. Since F_D^σ differs from F_D^χ by a constant, $\Psi_{M_\chi}(D)^p = \chi\Psi_{M_\chi}(D)$. Substitution into (11), using $M_\chi \equiv 1 - \chi H \pmod p$, proves (4) when $p \nmid M_\chi$. Changing the sign of v_0 negates both r and $\Psi_{M_\chi}(D)$.

Hasse's bound gives $|\tau| \leq 2\sqrt{p} < p - 1$ for $p \geq 7$. Thus $p \mid M_\chi$ occurs precisely when $\tau = \chi$, and then $M_\chi = p$. The class of D is nonzero: distinct points on a genus-one curve cannot have principal difference. It therefore has order p. The regular identity loses its scalar in this case. The bound excludes $p = 5$, where traces ± 4 are possible, from this particular degeneracy argument.

Direct exceptional normalization

Assume $\tau = \chi \in \{1, -1\}$ and identify $E = (v^2 = f(w), P_+)$, so $O = P_+$ and $\omega = dw/v$. Work over $k = \overline{\mathbb{F}}_p$. For a degree-zero divisor D_0 with p-torsion class, choose $\operatorname{div}\big(F_{D_0}\big) = pD_0$. Its logarithmic differential is holomorphic and defines

$$dF_{D_0}/F_{D_0} = \Psi_p([D_0])\omega, \qquad \Psi_p\colon \operatorname{Pic}^0(E)[p](k) \to k. \tag{13}$$

A principal change $D_0 \mapsto D_0 + \operatorname{div}(g)$ changes the function by a constant times g^p and leaves its logarithmic differential unchanged. Multiplication of functions gives additivity. The map is injective by Serre's logarithmic-differential construction (Serre, 1958, Proposition 10). Since $\mathcal{C}\omega = \chi\omega$, its image lies in the one-dimensional $\mathbb{F}_p$-space

$$\mathcal{L}_\chi = \{b \in k\colon b^p = \chi b\}. \tag{14}$$

By the residue divisor of a differential we mean the formal sum of its residues at geometric points. When compared with an integer divisor, its coefficients are reduced in k, hence modulo p. In particular, logarithmic residues are integer multiplicities reduced modulo p.

Sublemma 3a (Separable pullback). Cartier commutes with pullback by a separable morphism of smooth curves. Indeed, in a separating coordinate x, a rational differential has a unique expression $\sum_{i=0}^{p-1} u_i^p\, x^i\, dx$, and Cartier selects $u_{p-1}\, dx$. A separable pullback keeps the coordinate separating, so applying the same expression after pullback proves the assertion.

We use the kernel differential and quasi-period construction in Voloch (1997): Θ_R in (18) corresponds to his differential $f_P\omega$, and (19) is its translation law. The proof below fixes the scale and residue divisor for both signs. The published discussion after Voloch's Proposition 3 prints a negative sign in the quasi-period/logarithmic-derivative formula. With $t_T(Q) = Q + T$ and divisor $p\big((T) - (O)\big)$, the divisor calculation in (17) gives the positive sign used here; the proof determines that sign directly.

Lemma 3 (Exceptional Cartier defect). Suppose $\tau = \chi \in \{1, -1\}$. Let D_0 be a degree-zero integer divisor whose class is killed by p and whose coefficient conjugate is $\sigma(D_0) = \chi D_0$ as divisors. If θ_{D_0} is defined over $\mathbb{F}_{p^2}$, has only simple poles with residue divisor D_0, and satisfies $\sigma\big(\theta_{D_0}\big) = \chi\theta_{D_0}$, then

$$(\mathcal{C} - \mathrm{id})\theta_{D_0} = \Psi_p([D_0])\omega. \qquad (15)$$

Equivalently, the Cartier-defect map and the logarithmic transfer are the same additive map from geometric p-torsion into $\mathcal{L}_\chi$; the normalization factor is exactly 1.

Proof. First consider $D_P = (P) - (O)$ for $\chi = 1$, or $D_P = (P) - (-P)$ for $\chi = -1$, where $P \neq O$ is geometric p-torsion. Let V be Verschiebung, identifying the Frobenius twist with E through its model over $\mathbb{F}_p$. The Frobenius relations and invariant differential give

$$\pi + V = [\chi], \qquad \pi V = [p], \qquad V^*\omega = \chi\omega, \qquad \mathcal{C}\omega = \chi\omega. \qquad (16)$$

The curve is ordinary, so V is separable and étale of degree p. Its kernel $K = \ker V(k)$ is the cyclic group of geometric p-torsion, and every $T \in K$ automatically satisfies $\pi(T) = \chi T$. Thus $K = E\big(\mathbb{F}_p\big)$ if $\chi = 1$; if $\chi = -1$, it is the group $\{T : \pi(T) = -T\}$ corresponding to the rational points on the anomalous quadratic twist.

The divisor $\Delta = \sum_{T\in K} (T) - p(O)$ is principal: its group sum is O, because nonzero points pair as $T, -T$. It is Galois stable. The one-dimensional Riemann–Roch space $L(-\Delta)$ is defined over $\mathbb{F}_p$, so it has a nonzero base-field element s with $\mathrm{div}(s) = \Delta$. Put $\xi = \mathrm{dlog}\, s$ and $\psi(T) = \Psi_p([(T) - (O)])$. With translation $t_T(Q) = Q + T$, permutation of the kernel points gives

$$\mathrm{div}\frac{t_T^* s}{s} = p(O) - p(-T), \qquad t_T^*\xi - \xi = \psi(T)\omega. \qquad (17)$$

The sign is positive because $[(O) - (-T)] = [(T) - (O)]$. Also $\mathcal{C}\xi = \xi$ and $\psi(T)^p = \chi\psi(T)$.

For $R \in E(k)$, put $Q = V(R)$. Equation (17) makes $t_{-R}^*\xi - \xi$ invariant under K. Division by $V^*\omega$ gives a K-invariant rational function, which descends through the separable quotient V. Hence there is a unique rational differential Θ_R on the target with

$$\begin{aligned} V^*\Theta_R &= t_{-R}^*\xi - \xi = \mathrm{dlog}\frac{t_{-R}^* s}{s}, \\ \mathrm{div}\frac{t_{-R}^* s}{s} &= V^*\big((Q) - (O)\big) - p\big((R) - (O)\big). \end{aligned} \qquad (18)$$

As V is étale, Θ_R has only simple poles and residue divisor $(Q) - (O)$; the terms divisible by p have zero logarithmic residues. Sublemma 3a and injectivity of separable pullback give $\mathcal{C}\Theta_R = \Theta_R$. Translation and coefficient conjugation give

$$\Theta_{R+T} = \Theta_R - \chi\psi(T)\omega \quad (T \in K), \qquad \sigma(\Theta_R) = \Theta_{\pi(R)}. \qquad (19)$$

Choose R with $V(R) = P$, so $\pi(R) = \chi R - P$. For $\chi = 1$, set $\alpha = \Theta_R$ and $\beta = \psi(P)$; for $\chi = -1$, set $\alpha = \Theta_R - \Theta_{-R}$ and $\beta = 2\psi(P)$. In each case α has residue divisor D_P and $\beta = \Psi_p([D_P])$. The factor 2 is required because $[(P) - (-P)] = 2P$. Equation (19) gives

$$\mathcal{C}\alpha = \alpha, \qquad \sigma(\alpha) = \chi\alpha + \chi\beta\omega, \qquad \beta^p = \chi\beta. \qquad (20)$$

In the negative case, explicitly, $\sigma(\alpha) = \Theta_{-R-P} - \Theta_{R+P} = -\alpha - 2\psi(P)\omega$. A prescribed eigen differential θ_P differs from α by $c\omega$. Its eigencondition gives

$$c^p - \chi c = -\chi\beta, \qquad \chi c^{1/p} - c = \beta. \qquad (21)$$

Thus $(\mathcal{C} - \mathrm{id})\theta_P = \beta\omega$.

To pass to D_0, represent its class by D_P, choosing half the class when $\chi = -1$; multiplication by 2 is invertible on the group of order p. The zero class is handled by $\mathrm{dlog} g$ directly. Choose a third-kind differential η with residues D_P over $\mathbb{F}_{p^2}$ and replace it by $(\eta + \chi\sigma\eta)/2$ to satisfy the eigencondition. Conversely, the eigencondition implies descent to $\mathbb{F}_{p^2}$ because $\sigma^2\theta = \theta$. Write $D_0 - D_P = \mathrm{div}(g)$. Its eigencondition implies $g^\sigma = cg^\chi$ for some $c \in k^\times$, so $\mathrm{dlog} g$ is an eigen differential. The difference $\theta_{D_0} - \theta_P - \mathrm{dlog} g = b\omega$ has $b^p = \chi b$, hence is Cartier fixed. Cartier also fixes $\mathrm{dlog} g$. This proves (15).

This reduction shows that the defect depends only on the divisor class: changing an admissible differential adds a Cartier-fixed $b\omega$, and a principal eigen divisor contributes $\mathrm{dlog} g$, also Cartier fixed. The map is additive. Its image lies in $\mathcal{L}_\chi$, as does the injective transfer (13). The proof determines their comparison scalar as 1, including the factor of 2 in the negative eigencase. The kernel function and the preimage under V are proof constructions and are not computed by the evaluator.

Completion of Theorem 1. In the exceptional case put $\hat{\theta} = -\theta_\lambda / r$. It has residue divisor D and coefficient eigenvalue χ. Lemma 3 gives $(\mathcal{C} - \mathrm{id})\hat{\theta} = \Psi_p(D)\omega$. Since $r^{1/p} = \chi r$, evaluate at O to obtain

$$S_f(\lambda)^{1/p} = \chi - \chi r \Psi_p(D), \qquad \left(\chi r \Psi_p(D)\right)^p = \chi r \Psi_p(D). \qquad (22)$$

Raising to the pth power proves (4) with $H = \chi$.

For concrete exceptional examples, direct expansion and the normalized Miller evaluator give the following values in $\mathbb{F}_{11}$, with $\lambda = 1$ and origin $(0,1)$:

Table 2. Exceptional examples over $\mathbb{F}_{11}$.

$f(w)$	τ	$d = f(1)$	χ	H	$S_f(1)$	$r\Psi_p(D)$
$1 + w + 3w^3$	1	5	1	1	9	3
$1 + w + 2w^2 + 3w^3$	-1	7	-1	10	0	1

Remark on the high part. Put $\mathcal{G}_f(\mu) = \sum_{m \geq p} c_m \mu^m$. The Euler-criterion identity after (2) and Theorem 1 give

$$\mathcal{G}_f(\mu) = \chi - H + \chi r \Psi_{M_\chi}(D). \qquad (23)$$

In the exceptional case $H = \chi$, this reduces to $\mathcal{G}_f(\mu) = \chi r \Psi_p(D)$. For a trace 1 short Weierstrass curve $y^2 = F(x)$, with origin at infinity and $\omega_0 = dx/(2y)$, the corresponding classical form is

$$F(x)^{(p-1)/2} = \mathcal{U}(x) + x^{p-1} + x^p \mathcal{V}(x), \qquad y\mathcal{V}(x) = \Psi_p([(P) - (O)]), \quad P = (x, y). \qquad (24)$$

Here $\deg \mathcal{U} \leq p - 2$; the polynomial $\mathcal{V}$ is distinct from Verschiebung V. Its product with y is the Artin–Schreier descent function, and the right side is the logarithmic transfer (Deuring, 1941; Cassels, 1971; Voloch, 1990, n.d.; Semaev, 1998).

The Cartier calculation underlying this normalization is

$$(\mathcal{C} - \mathrm{id})\frac{y+b}{x-a}\omega_0 = b\mathcal{V}(a)\omega_0, \qquad P = (a,b). \qquad (25)$$

It is proved for every trace in Lemma 6 below, which also derives $b\mathcal{V}(a) = \alpha_{\#E(\mathbb{F}_p)}(P)$. At trace one, $\alpha_p(P)\omega_0 = d\log f_{p,P}$, so the sign and scale agree with (24) and with Lemma 3. The representative $(P) - (-P)$ has twice this scalar. At O, set the map to zero.

Relation between the two exceptional signs

For a trace -1 curve, the quadratic twist has trace 1. On geometric p-torsion, the negative Frobenius eigenspace becomes the rational eigenspace of the twist, so the underlying logarithmic transfer is the twisted trace-1 construction. Transporting that map still requires the chosen differential scale and the specified residue divisor. In short coordinates the divisor $(P) - (-P)$ represents twice $(P) - (O)$, which accounts for the factor 2. Thus twisting explains the relation between the signs, while Lemma 3 records the normalization for the actual two-pole representative used in Theorem 1.

Computing the logarithmic derivative

Use the quartic-to-Weierstrass construction with the rational point as origin (Connell, 1999, Section 1.2); the following formulas specify the version and scale required here. Write $f = 1 + f_1 w + f_2 w^2 + f_3 w^3 + f_4 w^4$, with $f_4 = 0$ for a cubic, and set

$$x = \frac{2(v+1) + f_1 w}{w^2}, \qquad y = \frac{4(v+1) + 2(f_1 w + f_2 w^2) - f_1^2 w^2/2}{w^3}. \qquad (26)$$

Substitution gives the generalized Weierstrass equation with

$$a_1 = f_1, \quad a_2 = f_2 - f_1^2/4, \quad a_3 = 2f_3, \quad a_4 = -4f_4, \quad a_6 = a_2 a_4. \qquad (27)$$

Near P_+, direct expansion gives

$$\begin{aligned} x &= 4w^{-2} + 2f_1 w^{-1} + f_2 - f_1^2/4 + O(w), \\ y &= 8w^{-3} + 4f_1 w^{-2} + (4f_2 - f_1^2)w^{-1} + O(1). \end{aligned} \qquad (28)$$

Thus P_+ maps to the identity at infinity. The points at infinity on the quartic map to affine points on this model; the finite points $Q_\pm$ also have affine images $R_\pm \neq O$, since $1/\lambda \neq 0$.

Put $b_2 = a_1^2 + 4a_2$, $b_4 = 2a_4 + a_1 a_3$, $b_6 = a_3^2 + 4a_6$, and use $X = x + b_2/12$, $Y = y + (a_1 x + a_3)/2$. The short equation is $Y^2 = X^3 + AX + B$, where

$$A = b_4/2 - b_2^2/48, \qquad B = b_6/4 - b_2 b_4/24 + b_2^3/864. \qquad (29)$$

All denominators are units for $p \geq 7$. The invariant differentials are proportional; comparing their leading terms in (28) fixes

$$\omega = \kappa\frac{dX}{2Y}, \qquad \kappa = -2. \qquad (30)$$

For a point R on the short model, use Miller functions and their local constants

$$\begin{aligned} \mathrm{div}(f_{m,R}) &= m(R) - ([m]R) - (m-1)(O), \\ \alpha_m(R) &= [z^0]\frac{\mathrm{dlog} f_{m,R}}{dz}, \quad z = -X/Y. \end{aligned} \qquad (31)$$

This is a Laurent coefficient of df/f, not an analytic logarithm, and multiplication of a Miller function by a scalar does not change it. The divisor recurrence is standard (Miller, 2004; Galbraith, 2012, Section 26.3.1). A nonvertical chord or tangent of slope ρ and a vertical line have expansions

$$\begin{aligned} Y - \rho X - \nu &= -z^{-3}\big(1 + \rho z + O(z^3)\big), \\ X - c &= z^{-2}\big(1 - cz^2 + O(z^4)\big). \end{aligned} \qquad (32)$$

The constant logarithmic derivative is therefore $+\rho$ for the nonvertical line and 0 for the vertical line. Binary doubling and addition update the accumulator by

$$\alpha \leftarrow 2\alpha + \rho, \qquad \alpha \leftarrow \alpha + \rho, \qquad (33)$$

respectively. Vertical and identity steps contribute zero. Doubling the accumulator is still necessary when the current point is O.

Since $[M_\chi]R_+ = [M_\chi]R_-$, the quotient $f_{M_\chi,R_+}/f_{M_\chi,R_-}$ has divisor $M_\chi D$. Its pole terms at O cancel, and $dX/(2Y) = \big(1 + O(z^4)\big)\,dz$, so

$$\Psi_{M_\chi}(D) = \frac{\alpha_{M_\chi}(R_+) - \alpha_{M_\chi}(R_-)}{\kappa}. \qquad (34)$$

Compute without finding a square root in the two-dimensional étale algebra $\mathcal{A}_d = \mathbb{F}_p[s]/(s^2 - d)$. Run the chain with $v_0 = s$ and write its constant as $\alpha = A_0 + B_0 s$. Algebra conjugation gives the other chain. Equations (4) and (34) become

$$r\Psi_{M_\chi}(D) = \frac{2B_0}{\kappa\lambda} = -\frac{B_0}{\lambda}, \qquad S_f(\lambda) = H + \chi\frac{B_0}{\lambda}. \qquad (35)$$

For $\chi = -1$, the algebra is a field; for $\chi = 1$, it is a product of two fields. The latter requires branch tests componentwise. An element $u = A_0 + B_0 s$ with norm zero and $u \neq 0$ has $B_0 \neq 0$ and supplies the root $(A_0/B_0)^2 = d$. Split the saved state by $s \mapsto \pm A_0/B_0$ and resume the remaining binary steps in both fields. At addition, first handle an identity point; for equal abscissae use a vertical step when ordinates are opposite, and a tangent when the points coincide, treating an ordinate of zero as vertical. Otherwise use the chord. At doubling, an identity or ordinate-zero point is a zero-contribution step; otherwise use the tangent. A nonzero zero divisor in an equality test can also split the state before choosing these branches, so a tangent in one component and a vertical addition in the other are handled separately. Splitting can occur only once. This is the degree-two instance of dynamic evaluation, or the D5 principle (Della Dora et al., 1985).

Algorithm 1 (Differentiated Miller chain). The following pseudocode uses the short model and computes the constant in (31). Prepare the operation stream from the binary expansion of M, omitting its leading 1: each remaining bit contributes a doubling operation followed by an addition operation if the bit is 1. The stored operation index is the next uncommitted operation.

```
T := R; alpha := 0; j := 0
while j < length(operations):
   op := operations[j]
   Q := T if op is doubling, otherwise R
   attempt (Tnew, rho) := ADD(T, Q)
   if a nonzero zero divisor a + b*s is found:
      split saved R, T, alpha by s → +a/b and -a/b
      resume both field components at this same j; return both constants
   anew := 2*alpha + rho if doubling, otherwise alpha + rho
   (T, alpha, j) := (Tnew, anew, j + 1)
return alpha, T
```

The subroutine ADD returns $(Q, 0)$ if $T = O$, and $(T, 0)$ if $Q = O$. For $T = (X_1, Y_1)$ and $Q = (X_2, Y_2)$, it returns $(O, 0)$ if $X_1 = X_2$ and $Y_1 + Y_2 = 0$; this includes doubling a point of order 2. If $T = Q$ otherwise, use the tangent slope $(3X_1^2 + A)/(2Y_1)$; if $X_1 \neq X_2$, use the chord slope $(Y_2 - Y_1)/(X_2 - X_1)$. Then set $X_3 = \rho^2 - X_1 - X_2$ and $Y_3 = \rho(X_1 - X_3) - Y_1$, and return $\big((X_3, Y_3), \rho\big)$. If $X_1 = X_2$ but the tangent and vertical tests disagree between components, $Y_1 - Y_2$ supplies the splitting witness. Any attempted inverse of a nonzero element of norm zero also supplies a witness. Its coefficient b is nonzero and $(a/b)^2 = d$. The split uses the saved state before the failed operation, so no accumulator update is repeated. In either component, identity and vertical steps still apply the doubling update when required. Finally use (34), or (35) before a split, to recover the coefficient sum.

For fixed f and p, cache the model, exact trace, H, and M_χ ($\chi \in \{1,-1\}$). Character computation and the affine Miller chain use $O(\log p)$ base-field multiplications and inversions. Resuming at most two components preserves this bound. The streaming query needs a constant number of working field elements and $O(\log p)$ control bits; an implementation retaining chain states may instead use $O(\log p)$ field elements.

Corollary 4 (Deterministic polynomial-time evaluation). Given $p \geq 7$, a squarefree cubic or quartic $f \in \mathbb{F}_p[w]$ with $f(0) \neq 0$, and an admissible λ, $S_f(\lambda)$ is computable deterministically in time polynomial in $\log p$. For N_{qry} queries on the prepared curve, classical arithmetic gives

$$T_{\mathrm{total}} = T_{\mathrm{Schoof}}(p) + O\big(N_{\mathrm{qry}}(\log p)^3\big), \qquad T_{\mathrm{Schoof}}(p) = O((\log p)^9). \tag{36}$$

The stated setup bound is the conservative bound of Schoof's original algorithm, not an optimal point-counting bound. After setup, a query uses $O(\log p)$ field operations and at most $O(\log p)$ field elements plus $O(\log p)$ control bits. This query-space bound does not include the polynomial working space of point counting.

Proof. If $c_0 = f(0) \neq 1$, normalize $g = f/c_0$ and restore $S_f = c_0^h S_g$. Use the trace of $v^2 = g(w)$, which is a quadratic twist of the original curve when c_0 is nonsquare. Fixed-degree squarefreeness tests and the model change are polynomial in $\log p$. Schoof's deterministic point count gives the trace and annihilators (Schoof, 1985; see Schoof, 1995 for the survey). Theorem 1 includes exceptional traces. All query algebra has fixed dimension; a zero divisor supplies its own splitting root. Since $M_\chi = O(p)$, the chains have $O(\log p)$ length. With $n = \lceil \log_2 p \rceil$ and modular multiplication and inversion bit costs $M(n), I(n)$, the query costs $O\big(n(M(n) + I(n) + n)\big)$ bit operations. Classical arithmetic gives $O(n^3)$, proving (36); fast arithmetic gives $\tilde{O}(n^2)$. At $\lambda = 0$ use the trace residue of the normalized curve and restore c_0^h.

Practical trace preparation can use SEA or an automatic exact routine such as PARI/GP's `ellap`, with its available SEA data at large characteristic (PARI Group, n.d.). The exact trace can be cached once per curve and prime; a change of curve may require a new trace. These practical choices do not replace Schoof in the deterministic worst-case proof. The quarter-point appendix uses specialized Cornacchia/Gauss preparation. The separate character-sum experiment uses PARI 2.17.2 without SEA data: its recorded paths are CM formulas where applicable and baby-step/giant-step point counting on the general inputs. Neither experiment benchmarks a Schoof- or SEA-based implementation of this corollary. For many queries, computing and sharing a full coefficient row can still be preferable.

Application to the original quarter-point sum

For an odd prime p, put $h = (p-1)/2$ and define the rational sequence

$$t_0 = 1, \qquad t_{i+1} = t_i \frac{h+2+i}{2i+2}, \qquad U_p(L) = \sum_{i=0}^{L-1} (h+1+i)\, t_i. \tag{37}$$

Reduction is well defined for $0 \leq L \leq h$, because the denominators use powers of two and factorials below p. Put $a_i = \binom{2i}{i} 8^{-i}$ and $B_L = \sum_{i<L} a_i$. Upper negation gives, in $\mathbb{F}_p$,

$$t_i = 2^{-i}\binom{h+1+i}{i} \equiv 2^{-i}(-1)^i \binom{-3/2}{i} = (2i+1)\binom{2i}{i} 8^{-i}. \tag{38}$$

The ratio $a_{i+1}/a_i = (2i+1)/\big(4(i+1)\big)$ gives the telescoping certificate

$$((2i+1)^2 - 8)a_i = -4(i+1)(2i+7)a_{i+1} + 4i(2i+5)a_i. \tag{39}$$

Consequently,

$$U_p(L) \equiv 4B_L - 2L(2L+5)a_L \pmod{p}. \tag{40}$$

Both the prefix and the boundary coefficient are required for this weighted sum. Now take $p = 4n+1 \geq 13$, so $h = 2n$. Since $\binom{h}{i}(-1/2)^i = a_i$ in $\mathbb{F}_p$, B_n is a strict midpoint truncation of the binomial row of length $h+1$, although $n = (p-1)/4$ is a quarter-point relative to p. The full row sum is $(1/2)^h = (2/p)$.

Set $f(w) = 1 - w^4/2$. Its coefficient of w^{4i} in f^h is a_i, so $H = a_n$ and $S_f(1) = B_n + a_n$. Theorem 1 and (40) give

$$B_n = -\chi r \Psi_{M_\chi}(D), \qquad U_p(n) \equiv 4B_n + \frac{9}{4}a_n \pmod{p}, \qquad (41)$$

where $\chi = (2/p)$ and $r = 1/v_0$, $v_0^2 = 1/2$. The curve has $j = 1728$ and rational nonzero two-torsion, so its trace is even and the exceptional traces ± 1 do not occur in this particular application.

The trace is classical complex-multiplication data. With $p = u^2 + s^2$ and the odd coordinate normalized by $u \equiv 1 \pmod 4$, Gauss's congruence is

$$\binom{2n}{n} \equiv 2u \pmod{p}, \qquad a_n \equiv 2u\, 8^{-n} \pmod{p}. \qquad (42)$$

This convention agrees with Cosgrave and Dilcher (2010, Theorem 1). For $p \geq 17$, Hasses bound $2\sqrt{p} < p/2$ makes the trace the representative of a_n in $(-p/2, p/2)$; the remaining prime $p = 13$ has trace 4, checked directly. Cornacchia preparation and exponentiation supply the boundary and trace for a Miller evaluation of the prefix. The benchmarked preparation includes a sequential nonresidue scan. Corollary 4 gives the deterministic polynomial-time alternative by computing the exact trace, using $a_n = H$, and evaluating $B_n = S_f(1) - H$. At $p = 97$, $n = 24$, $B_n = 33$, $a_n = 79$ and $U_p(n) = 43$.

The congruence literature for truncated central-binomial sums includes Sun (2011). The example here makes no separate priority claim for its binomial consequences. The elliptic-Fermat-quotient interpretation is established in the anomalous setting of Satoh and Araki (1998, 1999); applying that terminology to the regular $\Psi_{M_\chi}(D)$ in this example is only an analogy, since it retains a specified divisor and annihilator.

Rounded-third prefix and boundary

For primes $p \geq 11$ with $p \equiv 2 \pmod 3$, set $m = (p-2)/3$ and $L = m + 1 = (p+1)/3$. Retain the definitions of a_i, the strict prefix B_L and $U_p(L)$ from (37)–(40). Here the prefix is another direct application of Theorem 1: take $f(w) = 1 - w^3/2$ and $\lambda = 1$. The coefficients of f^h at w^{3i} are a_i, and $3m < p \leq 3L$. Thus $S_f(1) = B_L$. The coefficient H vanishes because $3 \nmid p - 1$; the exact trace is zero by the trace congruence and Hasse's bound. Since $f(1) = 1/2 \neq 0$, the argument is admissible, and the differentiated Miller query evaluates B_L in $O(\log p)$ field operations, without a point-counting search in this family.

The boundary a_L needed in (40) has a different arithmetic interpretation. Put $G = m! \mod p$. Wilson reflection and the ratio of consecutive terms give

$$a_L = \frac{3}{2G^3}, \qquad G = \left(\frac{3}{2a_L}\right)^{(2p-1)/3} \quad \text{in } \mathbb{F}_p. \qquad (43)$$

Indeed, $(2m)!\, L! = -1$ and $L! = LG$ give $(2m)! = -3G^{-1}$. Also $8^m = 1/2$, so $a_m = -6G^{-3}$, while $a_L/a_m = -1/4$. Cubing is bijective on $\mathbb{F}_p^\times$, yielding the inverse formula. Thus either a boundary backend or a modular-factorial backend supplies the other with $O(\log p)$ additional field operations.

The cubic above becomes $E: y^2 = x^3 + 1/4$ under $x = -w/2$, $y = v/2$. For the following cohomological statement use the point at infinity O as origin and fix $\omega = dx/y$; this differential has the opposite sign to dw/v under that coordinate change. Write $F = x^3 + 1/4$ and $F^h = \mathcal{T} + x^p \mathcal{S}$, with $\deg \mathcal{T} < p$. Let U be E with O removed, and W be E with $(0,1/2)$ and $(0,-1/2)$ removed. A Čech–de Rham cocycle is (α_U, α_W, g) with $\alpha_U - \alpha_W = dg$, and a coboundary is $(du_U, du_W, u_U - u_W)$. Fix

$$e_1 = [(\omega, \omega, 0)], \qquad e_2 = \left[\left(x\omega, \frac{\omega}{2x^2}, \frac{2y}{x}\right)\right]. \qquad (44)$$

Proposition 5 (Normalized rounded-third boundary). The classes (e_1, e_2) form a basis of $H^1_{\mathrm{dR}}(E/\mathbb{F}_p)$. With the Cartier convention of (8),

$$d(y\mathcal{S}) = a_L\omega, \qquad \mathcal{C}(\omega) = 0, \qquad \mathcal{C}(x\omega) = \mathcal{C}\left(\frac{\omega}{2x^2}\right) = 2a_L\omega. \qquad (45)$$

Define $V_{\mathcal{C}}[(\alpha_U, \alpha_W, g)] = [(\mathcal{C}\alpha_U, \mathcal{C}\alpha_W, 0)]$ and $F_{\mathrm{abs}}[(\alpha_U, \alpha_W, g)] = [(0,0, g^p)]$. These maps are linear over $\mathbb{F}_p$. With columns giving images of basis vectors,

$$[V_{\mathcal{C}}] = \begin{pmatrix} 0 & 2a_L \\ 0 & 0 \end{pmatrix}, \qquad [F_{\mathrm{abs}}] = \begin{pmatrix} 0 & -2a_L \\ 0 & 0 \end{pmatrix}. \qquad (46)$$

Proof. For a polynomial P, put $\mathcal{D}(P) = FP' + F'P/2$, so $d(yP) = \mathcal{D}(P)\omega$. Since $h + 1/2 = 0$ in the field, $\mathcal{D}(\mathcal{T}) = -x^p\mathcal{D}(\mathcal{S})$. The expansion of F^h has exponents divisible by three, with $\deg\mathcal{T} = p - 2$, so $\deg\mathcal{D}(\mathcal{T}) \leq p$. Consequently $\mathcal{D}(\mathcal{S})$ is constant, equal to $[x^{p+1}]F^h/4 = a_L$. Cartier coefficient selection gives $[x^{p-1}]F^h = 0$, $[x^{p-2}]F^h = 2a_L$ and $[x^{p+1}]F^h = 4a_L$, proving (45).

The identity $d(2y/x) = \left(x - 1/(2x^2)\right)\omega$ verifies the second cocycle. Its transition function has a simple pole at O. A Čech coboundary would require a function regular away from O with just that simple pole, excluded by genus-one Riemann–Roch. Together with the nonzero regular differential class, it therefore gives a basis. Applying Cartier proves the first matrix. For the second, write $(2y/x)^p = u_U - u_W$ with $u_U = 2y\mathcal{S}$ and $u_W = -2y\mathcal{T}/x^p$. These functions are regular on their respective opens, and $du_U = du_W = 2a_L\omega$. Subtracting their coboundary from $(0,0,(2y/x)^p)$ gives $-2a_Le_1$. This proves (46). The companion extension supplies the regularity and basis arguments in full and states the Frobenius-twist conventions over perfect extensions (Oda, 1969).

The usual genus-one Cartier–Manin matrix on regular differentials is the single entry zero. The nonzero entry $2a_L = 3G^{-3}$ in (46) belongs to full de Rham cohomology in this specified basis. Although $a_L\omega$ is exact in the function field, it is nonzero in projective de Rham cohomology: its rational primitive has poles and cannot be made globally regular. This is a global regularity obstruction, not a prohibition on cancelling a pole at one point. Neither the zero trace nor this interpretation provides a fast boundary extractor, and no identification with a normalized Chowla–Selberg period is used.

Complete rounded-third evaluation

The boundary in (43) also supplies the adjacent cutoff $m = L - 1$. The consecutive-term ratio and (40) give

$$a_m = -4a_L, \qquad B_m = B_L + 4a_L, \qquad (47)$$

$$U_p(L) = 4B_L - \frac{17}{3}G^{-3}, \qquad U_p(m) = 4B_L - \frac{16}{3}G^{-3}. \qquad (48)$$

In particular, $U_p(L) - U_p(m) = -G^{-3}/3 \neq 0$, so the adjacent sums cannot both vanish. Here $G \neq 0$ because $m < p$. The boundary multiplier $2L(2L + 5) = 34/9$ vanishes only at $p = 17$ in this domain. Although the weighted sum then simplifies, the implementation still acquires and checks a_L and G separately.

The integrated single-prime routine validates p, obtains B_L from the unchanged differentiated Miller provider, acquires the signed boundary a_L by the selected backend, and assembles (40). Formula (43) recovers G when the backend supplies a_L directly; (48) then supplies the adjacent outputs. Apart from boundary acquisition, the complete routine requires $O(\log p)$ field operations, including modular powering. It returns the same tuple $\left(B_L, a_L, U_p(L), G\right)$ for every backend. The implemented public interface checks primality exactly in the supported domain $11 \leq p < 2^{63}$, $p \equiv 5 \pmod 6$.

Certified Jacobi boundary acquisition

Write $k_\ell = (p - 1)/\ell$ for a prime conductor $\ell \mid p - 1$. The bounded backend, called T3 in the code, supports primes $5 \leq \ell \leq 89$, with $k_\ell \geq 4$ required for forced calls. It reconstructs a Jacobi sum in $\mathbb{Z}[\zeta_\ell]$ by lattice reduction, accepts it only after the algebraic and normalization certificates, and evaluates its Galois conjugates to obtain grid-factorial relations. In particular, the norm relation $J\bar{J} = p$ is combined with the required root-of-unity normalization; a norm check alone is not the full certificate. Integer relation plans are checked on use. Two interval products connect the grid to G and to a rounded-third anchor. The interval lengths are approximately $k_\ell/3$ and $k_\ell/6$, or $k_\ell/3$ and $k_\ell/12$ when the denominator-12 anchor is available for $p \equiv$

$5 \pmod{12}$. The companion gives the exact residue-class formulas, relation plans, normalizations and certificates (Jordan, 2026a).

The denominator-12 anchor means an auxiliary factorial at $N = \lfloor (p-5)/12 \rfloor$ or $5N+2$, corresponding to rational fractions with denominator 12. It does not assert $12 \mid p-1$; only the prime Jacobi conductor ℓ must divide $p-1$. The saving lies in the anchor interval: the rounded-third relations replace the longer grid normalization by a shorter anchor tied to the same target. The signed half-factorial is not required, and neither class numbers nor GRH enter this integrated route. These are bounded specializations of Jacobi and modular-factorial methods, not a new general complexity theorem. For fixed ℓ, the interval work retains square-root dependence on p up to logarithmic and multiplication factors. Lattice reconstruction is not proved to succeed on every input; exact acceptance and guaranteed reconstruction are different claims.

The automatic policy uses native FLINT factorial below 10^7. Above that threshold it ranks eligible conductors with $k_\ell \geq 10^4$ by a frozen cost model and attempts T3 when its prediction beats the nodal alternative. Certified reconstruction misses are recorded; automatic mode tries the remaining candidates and then falls back to the exact nodal backend. Forced T3 raises an error if reconstruction cannot be certified. The threshold is an implementation policy, not a universal crossover. The nodal backend is the companion's factorial-ratio anchor evaluated by FLINT block products; it uses no class number. Native factorial remains an explicit comparator and alternative. This fallback policy preserves the mathematical output but does not guarantee the fastest route on every machine.

Remainder forests for dense prime tables

A remainder forest shares ordered integer products and their reductions across many primes. Product trees combine recurrence intervals and prime moduli; remainders descend to the leaves to recover the required product at each prime's cutoff. Processing successive smaller trees, with a carry reduced modulo the unprocessed primes, controls the size of the retained arithmetic state. This is established accumulating remainder-forest machinery (Harvey & Sutherland, 2016), specialized here to an upper-triangular recurrence.

Over the rationals put $\Delta_j = 4^j j!$, $\hat{B}_j = \Delta_j B_j$ and $\hat{a}_j = \Delta_j a_j$. Then

$$\mathsf{A}_j = \begin{pmatrix} 4(j+1) & 4(j+1) \\ 0 & 2j+1 \end{pmatrix}, \qquad \mathsf{A}_{j-1} \cdots \mathsf{A}_0 = \begin{pmatrix} \Delta_j & \hat{B}_j \\ 0 & \hat{a}_j \end{pmatrix}. \qquad (49)$$

This follows from $B_{j+1} = B_j + a_j$ and the ratio preceding (39); the empty product is the identity. At $j = L < p$, division by the nonzero Δ_L recovers B_L and a_L, while $G = \Delta_L/(4^L L)$ and (40) recover the other outputs. Thus the forest computes the complete tuple jointly and bypasses the Miller pass. Its gain is batch amortization, not an acceleration of an isolated Miller query. The companion provides ordered-product and carry invariants, bit-size and storage bounds, and reproduction commands (Jordan, 2026b, 2026a).

Dense prime schedules share enough recurrence work to benefit strongly. A sparse schedule still pays for products from index zero to its last requested cutoff, so independent calls can be preferable. The experiments compare these workloads separately. The integrated single-prime adapter does not automatically replace the workbench's existing many-prime dispatcher.

Supported evaluation modes

Table 3. Supported evaluation modes. The table distinguishes the geometric query from any additional boundary arithmetic.

Evaluation	**Prefix or coefficient sum**	**Additional arithmetic**
General admissible genus-one sum	Differentiated Miller query	Exact trace preparation
Original quarter-point weighted sum	Cartier–Miller prefix	Classical CM boundary and trace data
Rounded-third sum at one prime	Cartier–Miller prefix with known zero trace	Native factorial, nodal or certified Jacobi boundary
Rounded-third table across primes	Joint recurrence through a remainder forest	Shared prefix and boundary acquisition

For suitable individual primes the integrated Jacobi route improves complete evaluation, while dense tables favour the retained forest. Tables 5–7 give the matched rounded-third measurements; Table 8 gives a separate character-sum comparison.

Weighted character sums and elliptic-point sums

We now allow any squarefree cubic or quartic $f \in \mathbb{F}_p[x]$, without a restriction on its constant term. Put $\varepsilon_f(x) = \chi\big(f(x)\big)$, where $\chi(0) = 0$. All weighted sums in this section are elements of $\mathbb{F}_p$. Their output differs from an integer character sum, an incomplete sum over an interval, and the nonlinear expression $\sum_x \chi\left(R(x)f(x)\right)$. We first give an alternative single-pole evaluator using one rational-point Miller chain. The lemma also fixes the high-part normalization uniformly in the trace.

A uniform curve-order coefficient

Lemma 6 (Curve-order Miller coefficient). Let $p \geq 5$, let $E: y^2 = W(x) = x^3 + ux + v$ be nonsingular, and put $\omega_0 = dx/(2y)$, $t = -x/y$, $N = \#E\big(\mathbb{F}_p\big) = p + 1 - \tau_E$. Write

$$W(x)^{(p-1)/2} = U(x) + A_E x^{p-1} + x^p V_W(x), \qquad \deg U \leq p - 2. \qquad \text{(U1)}$$

With the Miller convention (31), for every affine rational point $P = (a, b)$,

$$\alpha_N(P) = bV_W(a). \qquad \text{(U2)}$$

There is no coprimality or ordinarity hypothesis. Set $\alpha_m(O) = 0$. For $p \nmid N$, the differential has a pole at O, so its Laurent constant is used; it is not its value at O, or a claimed additive embedding of the full rational point group.

Proof. Put $\kappa(S,T) = [t^0]\big(d\log g_{S,T}/dt\big)$, where $\operatorname{div} g_{S,T} = (S) + (T) - (S+T) - (O)$. A chord or tangent contributes its slope, a vertical line contributes zero, and an identity operand uses $g = 1$, as in (32)–(33). Comparing divisors proves

$$\begin{aligned} \alpha_m(S+T) &= \alpha_m(S) + \alpha_m(T) + \kappa([m]S, [m]T) - m\kappa(S,T), \\ \alpha_{mn}(Q) &= n\alpha_m(Q) + \alpha_n([m]Q). \end{aligned} \qquad \text{(U3)}$$

For the second identity use $f_{mn,Q} \doteq f_{m,Q}^{\,n} f_{n,[m]Q}$, where dotted equality allows a nonzero constant. These identities hold for geometric points, including identity endpoints. Negation gives $\alpha_m(-S) = -\alpha_m(S)$. In particular, α_p is additive when one summand is geometric p-torsion.

Suppose first $b \neq 0$. Choose a geometric Q with $[p]Q = P$ and set $\alpha = \alpha_p(Q)$. The differential $\vartheta_P = (y + b)\omega_0/(x - a)$ has residue divisor $(P) - (O)$. Since $W(a)^{(p-1)/2} = 1$, division of $W^{(p-1)/2} - 1$ by $x - a$ gives a polynomial of degree below $2p - 1$ whose x^{p-1} coefficient is $V_W(a)$. The Cartier coefficient rule, together with $1/y = W^{(p-1)/2}/y^p$, gives $(\mathcal{C} - 1)\vartheta_P = bV_W(a)\omega_0$. The differential $\theta = -d\log f_{p,Q}$ has the same residues and is Cartier fixed. At O, $\vartheta_P/\omega_0 = -t^{-1} + O(t)$ and $\theta/\omega_0 = -t^{-1} - \alpha + O(t)$. Thus $\vartheta_P - \theta = \alpha\omega_0$. Applying Cartier and raising to p, with $\mathcal{C}\omega_0 = A_E^{1/p}\omega_0$, proves

$$bV_W(a) = A_E\alpha - \alpha^p. \qquad \text{(U4)}$$

Let Φ be Frobenius and $\mathcal{V}$ Verschiebung. Since $\Phi\mathcal{V} = [p]$ and Frobenius is injective on geometric points, $\mathcal{V}Q = P$. The trace relation $\Phi + \mathcal{V} = [\tau_E]$ therefore gives $\Phi Q = Q - [N]Q$. Put $T = [N]Q$; then $[p]T = [N]P = O$. Galois compatibility and (U3) give

$$\alpha^p = \alpha_p(\Phi Q) = \alpha - \alpha_p(T), \qquad N\alpha + \alpha_N(P) = \alpha_{pN}(Q) = \alpha_p(T). \qquad \text{(U5)}$$

The last equality uses the composition identity in both orders and $p = 0$ in the coefficient field. As $A_E \equiv \tau_E \pmod p$, substitution in (U4) yields $bV_W(a) = (A_E - 1 + N)\alpha + \alpha_N(P) = \alpha_N(P)$. If $b = 0$, then N is even and $f_{N,P} \doteq (x - a)^{N/2}$; its logarithmic constant is zero, as is the right side of (U2). This proves the remaining case. The point Q is a proof device and is never computed.

At trace one, $N = p$, so (U2) is the positive normalization in (24). For the representative $(P) - (-P)$, the function $f_{N,P}/f_{N,-P}$ has logarithmic constant $2\alpha_N(P) = 2bV_W(a)$. This states explicitly the factor of two for that representative relative to ω_0 and t. When $p \nmid N$, choosing the rational point $Q = [p^{-1} \bmod N]P$ in (U4) also gives $bV_W(a) = (A_E - 1)\alpha_p(Q)$. Lemma 6 avoids this modular inverse and preliminary scalar multiplication. Its annihilator is the same curve order used in Theorem 1 after the appropriate twist.

The local-coefficient construction and its short-chain computation are classical (Semaev, 1998, Lemmas 1–3; Rück, 1999, p. 806). Gunji (1976, equations (2.4), (2.9), Lemma 2.11) supplies earlier Cartier and logarithmic-differential structure. Lemma 6 is derived here from those constructions and the Frobenius relation; publication priority for this combined scalar identity is not established.

One pole and a rational-point algorithm

Proposition 7 (Polynomial identity at every pole). For a squarefree cubic or quartic over $\mathbb{F}_p$, $p \geq 5$, write $F = f^{(p-1)/2} = L + x^p J_f$, $\deg L < p$. For every $a \in \mathbb{F}_p$, including branch values and zero,

$$C_f(a) := \sum_{x \neq a} \frac{\varepsilon_f(x)}{x - a} = -F'(a) - J_f(a). \qquad \text{(W1)}$$

Proof. Put $Q = \big(F - F(a)\big)/(x - a)$. The reciprocal sum over $x \neq a$ is zero, so $C_f(a) = \sum_x Q\,(x) - Q(a)$. Here $\deg Q \leq 2p - 3$; the power sums (10) give $\sum_x Q\,(x) = -[x^{p-1}]Q = -J_f(a)$, while $Q(a) = F'(a)$. At a nonbranch point, $F'(a) = -\varepsilon_f(a) f'(a)/\big(2f(a)\big)$. At a branch point, $F'(a) = 0$, since $(p-1)/2 \geq 2$. For $a \neq 0$, (W1) becomes $C_f(a) = \Big(L(a) - \varepsilon_f(a)\Big)/a - F'(a)$, recovering (51).

Theorem 8 (Uniform nonbranch query). Let $f(a) \neq 0$, set $e = \varepsilon_f(a)$, and put

$$g(z) = \frac{f(a+z)}{f(a)} = 1 + g_1 z + g_2 z^2 + g_3 z^3 + g_4 z^4, \quad g_1 = \frac{f'(a)}{f(a)}, \quad \tilde{\tau} = e\tau_f, \quad N = p + 1 - \tilde{\tau}. \qquad \text{(W2)}$$

Here τ_f is the exact projective trace and $g_4 = 0$ for a cubic. Apply (26)–(29) to $v^2 = g(z)$, taking $(0,1)$ as origin, and let R be the image of $(0, -1)$. On the resulting short model,

$$\boxed{C_f(a) = e\left(-\frac{g_1}{4}(N-2) - \frac{\alpha_N(R)}{2}\right).} \qquad \text{(W3)}$$

One base-field differentiated Miller chain evaluates this expression; it requires neither a square root nor a quadratic algebra.

Proof. Translation and normalization give $\varepsilon_f(a+z) = e\varepsilon_g(z)$ and $C_f(a) = eC_g(0)$. If $N_\infty(f)$ is the number of rational points at infinity, then $N_\infty(f) = 1$ for a cubic and $1 + \chi\big(\ell_f\big)$ for a quartic with leading coefficient ℓ_f. Fibre counting gives $\tau_f = 1 - N_\infty(f) - \sum_x \varepsilon_f\,(x)$. For a quartic normalization also changes the infinity character by e; this identity proves the signed integer relation $\tau_g = e\tau_f$ in both degrees.

The model in (26)–(29) has short coefficients and point

$$\begin{aligned} A_g &= -4g_4 + g_1 g_3 - g_2^2/3, \\ B_g &= g_3^2 - 8g_2 g_4/3 + g_1^2 g_4 - g_1 g_2 g_3/3 + 2g_2^3/27, \qquad \text{(W4)} \\ R = (r,s) &= (g_1^2/4 - 2g_2/3, -g_3 + g_1 g_2/2 - g_1^3/8). \end{aligned}$$

The formulas extend to an isomorphism of smooth projective curves. Substitution, or expansion at $(0,-1)$ followed by (26), gives R. An exact identity of rational functions is

$$\frac{1}{z} = \frac{1}{2} h_R - \frac{g_1}{4}, \qquad h_R = \frac{Y + s}{X - r}. \qquad \text{(W5)}$$

It follows directly from (26) and does not require division by s. Apparent singularities at removable points use their regular values. When $s \neq 0$, the value at $-R$ is $-W_g'(r)/(2s)$, where $W_g = X^3 + A_g X + B_g$. Summing the other fibres and applying (W1) on the short model gives

$$\sum_{\substack{S \in E_g(\mathbb{F}_p) \\ S \notin \{O,R\}}} h_R(S) = sC_{W_g}(r) - \frac{W_g'(r)}{2s} = -sV_{W_g}(r) = -\alpha_N(R). \qquad \text{(W6)}$$

If $s = 0$, negation pairs the terms $Y/(X - r)$; the fixed points contribute zero, and Lemma 6 gives $\alpha_N(R) = 0$. Thus the same identity holds for a nonzero rational point of order two.

The function $1/z$ vanishes at every point at infinity of the cubic or quartic model. Its affine point sum with $z \neq 0$ is $C_g(0)$, since $\sum_{z \neq 0} 1/z = 0$. The short-model sum excludes exactly O, R, hence has $N - 2$ terms. Summing (W5) and using (W6) proves (W3). The geometric construction in Lemma 6 supplies its sign for all traces; no exceptional constant is selected experimentally.

In particular, a zero pole and a zero constant term of the original f present no normalization obstruction when $f(a) \neq 0$. Every pole is used as its own normalization origin. The same cached original trace supplies each twist trace. Formula (W3) includes $p = 5$, even when $N = 2p$; for example it returns $C_{x^3+3x}(1) = 1$ over $\mathbb{F}_5$, with $\tau_f = -4$. This extends the original $p \geq 7$ formulation. Branch poles remain outside the fast query.

Several poles and complete excluded domains

Theorem 9 (Supplied simple-pole weights). Let $U = \{a_1, \ldots, a_s\} \subset \mathbb{F}_p$ be distinct, $D = \{x \in \mathbb{F}_p : x \notin U\}$, and let $R(x) = c + \sum_i r_i/(x - a_i)$ be supplied in partial fractions. Zero residues are allowed, so U records the entire excluded set, including extra exclusions. Put $w_i = \varepsilon_f(a_i)$, $K_f = \sum_x \varepsilon_f(x) = 1 - N_\infty(f) - \tau_f$, and $D_i(v) = \sum_{j \neq i} v_j/(a_i - a_j)$. Then

$$\begin{aligned} \sum_{x \in D} R(x)\varepsilon_f(x) &= c\left(K_f - \sum_i w_i\right) + \sum_i r_i\left(C_f(a_i) + D_i(w)\right), \\ \sum_{\substack{P \text{ affine} \\ x(P) \in D}} R\left(x(P)\right) &= c\left(K_f - s - \sum_i w_i\right) + \sum_i r_i\left(C_f(a_i) + D_i(1 + w)\right). \end{aligned} \qquad \text{(W7)}$$

The difference of the point sum and character sum is the elementary baseline $-sc + \sum_i r_i D_i(1)$. For one reciprocal pole and $c = 0$, it is zero. The two sums therefore express the same evaluation problem up to an explicitly computable baseline.

Proof. The oracle $C_f(a_i)$ excludes only a_i. Removing each other a_j subtracts $w_j/(a_j - a_i)$, giving the plus correction $D_i(w)$. The affine fibre over x has the integer size $1 + \varepsilon_f(x)$. Its reciprocal baseline is $\sum_{x \in D} (x - a_i)^{-1} = D_i(1)$; its constant baseline is $|D| = p - s \equiv -s \pmod p$. These facts prove both formulas and their difference. Corrections must use all of U, even where $r_i = 0$. The formulas also hold for D empty.

The weight tends to c at infinity. The projective rational-point sum, retaining all rational points at infinity, consequently adds $cN_\infty(f)$ to the affine sum. No character value at infinity is assigned. A positive-degree polynomial part instead has a pole there and needs a separately specified domain; a quartic with $N_\infty(f) = 0$ has no rational infinity point to exclude. For a general function $A(x) + yB(x)$, cancellation of the odd part by negation is valid only on whole included fibres where the function is regular, with removable singularities evaluated by their limits.

Query costs and admissibility

Corollary 10 (Field and bit bounds). Assume every active pole is nonbranch and the exact signed trace is supplied. With $n = \lceil \log_2 p \rceil$, Theorem 8 uses $O(n)$ field operations, including $O(n)$ inversions, and a constant number of field registers with $O(n)$ control bits. Writing $M(n), I(n)$ for multiplication and inversion bit costs,

a safe query bound is $O\big(n(M(n)+I(n)+n)\big)$. This is $O(n^3)$ classically and $\tilde{O}(n^2)$ with fast arithmetic. Trace acquisition and primality certification are separate input costs.

For (W7), character evaluation at all exclusions and schoolbook corrections give $O(sn+s^2)$ field operations. Batch inversion of the nonzero differences gives the bit bound

$$O\big(sn(M(n)+I(n)+n)+s^2M(n)+I(n)\big), \quad O(sn^3+s^2n^2) \text{ classically.} \qquad \text{(W8)}$$

An exclusion with zero residue may be a branch point, since its oracle value is not needed. No denominator factorization or unsupplied partial-fraction decomposition is included in these bounds.

Proof. Forming (W2), (W4) costs a constant number of field operations apart from the character exponentiation. By Hasse's bound $N = O(p)$, so the binary Miller chain has $O(n)$ steps. All denominators in the fixed-degree model change are units for $p \geq 5$. Identity and vertical steps are covered by (32)–(33), including an intermediate endpoint O. Multiplying s query costs and adding the pairwise corrections proves (W8). If $s = 0$, only the trace-derived constant term remains. Reading s arbitrary field inputs already costs $\Omega(sn)$ bits. The endpoint check from the chain is a necessary consistency check for a supplied trace, not a trace certificate.

For completeness, the correction vector has a fast polynomial construction. Set $B(X) = \prod_j \big(X - a_j\big)$ and $W_v(X) = \sum_j v_j\, B(X)/\big(X - a_j\big)$. Taylor expansion at a_i gives

$$D_i(v) = \frac{W_v'(a_i)}{B'(a_i)} - \frac{v_i B''(a_i)}{2B'(a_i)}. \qquad \text{(W9)}$$

Indeed the derivative of the ith summand is $v_i B''(a_i)/2$, and the remaining derivatives sum to $B'(a_i)D_i(v)$. A product tree constructs B and W_v: at a leaf use $(X - a_i, v_i)$, and combine disjoint children by $(B_L B_R, W_L B_R + W_R B_L)$. Derivative formation, remainder-tree evaluation and one batch inversion then cost $O\left(\mathsf{M}_{mpol}(s)\log(s+1)\right)$ field operations under the usual polynomial-arithmetic cost assumptions, plus one scalar inversion. This improves the correction term in (W8) accordingly, with storage for the polynomial trees. It is an arithmetic bound, not a new benchmark. For fixed curve and excluded set, caching the oracle values and correction vectors makes subsequent coefficient-only changes $O(s)$-operation dot products.

These results establish an accelerated class of constant-plus-simple-pole weights. They do not assert the same bound for every rational weight. At a branch pole (W1) remains valid, but its additional high coefficient has not been supplied by Theorem 8. Higher-order poles require further local coefficients or Hasse derivatives. A base-field pointwise identity such as (4) cannot be differentiated in a transcendental argument to obtain them. The trace-free selected-coefficient method below remains available at branch poles. Likewise, the extension-field and higher-genus problems in the next section require separate proofs.

A trace-free selected-coefficient comparator

Put $d = \deg f$, $h = (p-1)/2$, $G_a(z) = z^d f(a + 1/z)$, and $k = dh - p < p$. Since $(a+z)^p = a + z^p$ and $\deg J_f < p$, the coefficient of z^p in $F(a+z) = L(a+z) + (a+z^p)J_f(a+z)$ is $J_f(a)$. Reversing this polynomial proves

$$J_f(a) = [z^k]G_a(z)^h, \qquad G_a(0) = \ell_f \neq 0. \qquad \text{(W10)}$$

This includes zero and branch poles without trace preparation. If $G_a = \sum_{j=0}^{d} g_j\, z^j$ and $G_a^h = \sum e_k z^k$, then

$$k g_0 e_k = \sum_{j=1}^{\min(d,k)} \big((h+1)j - k\big)\, g_j e_{k-j}, \qquad e_0 = g_0^h. \qquad \text{(W11)}$$

All denominators are nonzero for $1 \leq k < p$. The state $k!\, g_0^k(e_k, \dots, e_{k-d+1})$ evolves through a fixed-size matrix linear in k: its top row is $\big((h+1)j - k\big)g_j$, and its subdiagonal entries are $k g_0$. Negative-index coefficients are zero. Streaming costs $O(p)$ field operations and constant storage.

Bostan–Gaudry–Schost block products (Bostan et al., 2007) and Harvey's interval-product theorem (Harvey, 2007, Theorem 10) give $\tilde{O}(\sqrt{p})$ field operations for fixed d, with $\tilde{O}(\sqrt{p})$ storage. Harvey's theorem concerns linear matrix polynomials, ordered interval products and suitable invertible interpolation factors. For the single interval $[1,k]$, its sufficient unit condition holds because, with $s_0 = \lfloor \log_4 k \rfloor$, $2^{s_0} + 1 < p$. The interpolation product and its inverse, and the scalar denominator $k!\, g_0^k$, must be acquired within the same square-root budget; they are not free preparation. Handle $k \le 1$ directly. The precise ring-operation bound is $O\left(\mathsf{MM}(d)\sqrt{k} + d^2 \mathsf{M}_{\mathrm{pol}}(\sqrt{k})\right)$, apart from the smaller initialization exponentiation. The interval product order must agree with the column-state recurrence. The literature supplies the block-product machinery; (W10) supplies the matched weighted-sum adapter. These trace-free costs and the trace-supplied cost in Corollary 10 describe different preparation models. Neither is an unconditional ranking for dense-query workloads.

Scope and an independent point sum

At a branch argument $f(1/\lambda) = 0$, the two simple poles of θ_λ merge into a residue-free pole of the second kind. Removing exact differentials leaves a second-kind cohomology class whose Cartier image involves an additional polynomial-power coefficient, beyond the trace. Katz's note (n.d.) illustrates this distinction through Cartier on dx/y and $x\,dx/y$. This observation does not establish a fast branch evaluator. The division-free streaming fallback takes $O(p)$ field operations. Equations (W10)–(W11) also supply a trace-free $\tilde{O}(\sqrt{p})$ selected-coefficient route; neither bound is the nonbranch logarithmic query of Theorem 8.

If $f(0) = 0$, squarefreeness gives $f = wg$ with $g(0) \neq 0$, and the coefficient problem becomes $[w^h]g(w)^h/(1-\lambda w)$ rather than the normalized cutoff $p-1$. The origin and local coefficient selection used above therefore change; this case is not included in Theorem 1.

Proposition 2 already holds over perfect extension fields. For $q_k = p^k$ and $\lambda \in \mathbb{F}_{q_k}^{\times}$, put $e_k = d^{(q_k-1)/2} \in \{1,-1\}$ and define $\tau_0 = 2$, $\tau_1 = \tau$, $\tau_k = \tau\tau_{k-1} - p\tau_{k-2}$. The divisor class is annihilated by $M = q_k + 1 - e_k\tau_k$. If $p \nmid M$, (11) gives $S_f(\lambda) = H + (r/M)(H\Psi_M(D) - \Psi_M(D)^p)$, with a Miller chain of length $O(k\log p)$. Since $\tau_k \equiv \tau^k \pmod p$, the condition $p \mid M$ is equivalent to $e_k\tau^k \equiv 1 \pmod p$. The base-field simplification and equality with $T_f(1/\lambda)$ require separate arguments over extensions.

An elementary verification identity gives an independent $O(p)$ computation for admissible base-field arguments. With $\mu = 1/\lambda$ and $d = f(\mu)$,

$$S_f(\lambda) = \chi - \frac{\chi f'(\mu)}{2d\lambda} - \sum_{\substack{P \in E(\mathbb{F}_p) \text{ affine} \\ w(P) \neq \mu}} \frac{1}{1-\lambda w(P)}. \qquad (50)$$

To check it, write $F = f^h$ and $Q = \big(F - F(\mu)\big)/(w-\mu)$. The integer number of affine points over w is congruent to $1 + F(w)$ modulo p. Summing after polynomial division, using (10) and $Q(\mu) = F'(\mu)$, yields the point sum $F(\mu) - T_f(\mu) + F'(\mu)/\lambda$. Since $F'(\mu) = -\chi f'(\mu)/(2d)$, this is (50). Points at infinity contribute zero if the summand is extended by its limiting value. This identity neither uses Miller functions nor proves their exceptional normalization by itself.

Define $\varepsilon_f(w) = \left(\frac{f(w)}{p}\right)$ first, with value zero at a root of f. The integer number of affine points above w is $1 + \varepsilon_f(w)$. Since $\sum_{w \neq \mu}(1-\lambda w)^{-1} = 0$ in $\mathbb{F}_p$, the same point-sum identity becomes

$$S_f(\lambda) = \chi - \frac{\chi f'(\mu)}{2d\lambda} + \mu \sum_{w \neq \mu} \frac{\varepsilon_f(w)}{w-\mu}. \qquad (51)$$

The scalar $\chi = \varepsilon_f(\mu)$ is distinct from the character function ε_f. Substituting Theorem 1 in this display proves Corollary 1.1.

Higher genus

The residue cancellation behind Proposition 2 extends formally to a smooth curve of genus g: the holomorphic remainder after cancellation of the simple poles is written in a holomorphic basis as $\eta = \sum_{i=1}^{g} c_i\, \omega_i$, and $\mathcal{C}\eta = \sum_i c_i^{1/p}\, \mathcal{C}\omega_i$ involves the Cartier–Manin matrix. For a principal multiple of a divisor, logarithmic differentials and Jacobian transfer maps remain available (Rück, 1999). For $g > 1$, evaluation at the distinguished point supplies only one linear functional on this vector correction. Replacing H by a matrix alone therefore does not give the coefficient-sum formula or its exceptional normalization; these require further choices and arguments. No higher-genus evaluator is asserted here.

Computational verification

Retained code compares the generic evaluator with direct polynomial powers and independent point counts. An audit over primes $11 \le p \le 499$ tested 20,316 admissible queries on 906 sampled squarefree cubics and quartics, including 368 exceptional queries and 2,512 split-algebra fallbacks, with no mismatches. The attribution revision reran this sample and additionally checked the high-part identity in the remark after Theorem 1 on every query; the character-sum form following (50) agreed on its 4,072 admissible queries with $p < 90$. Exhaustive compiled tests over $\mathbb{F}_5$ and $\mathbb{F}_7$ agreed with FLINT polynomial powers (Hart, 2010); the $p = 5$ checks are computational evidence outside Theorem 1's stated range; Theorem 8 separately covers these small-field nonbranch character queries.

A separate exhaustive check enumerated all 343 nonsingular short Weierstrass models $y^2 = x^3 + Ax + B$ of trace 1 for $7 \le p < 48$. At all 11,730 nonidentity rational points, direct expansion of the high part agreed with the independent Miller slope accumulator relative to $dx/(2y)$. The identity point was checked separately on each model, and all 463,615 ordered pairs of rational points satisfied additivity. The supplementary $p = 5$ check covered two models and eight nonidentity points, outside Theorem 1's prime range.

Separate retained direct-proof checks covered 17,940 nonzero torsion-point evaluations on 354 curves for $7 \le p \le 101$, and 568 kernel-translation checks on 32 curves for $7 \le p \le 31$. An independent point-sum verifier tested 4,357 queries on 384 curves for $7 \le p \le 103$, including 155 exceptional queries, with no mismatches. Source code, detailed logs and reproduction commands accompany the manuscript and the rounded-third companion extension. These finite checks do not replace proof review.

The earlier rounded-third companion has a separate validation record (Jordan, 2026b). Its anchor suite checked all 152 admissible primes below 2,000 and 595 reconstructed conductor values against direct character enumeration. Its native-FLINT comparison comprised 30 complete evaluations over five selected primes. Its million-bound standard-library batch had independent prefix checks, complete checks below 20,000 and 40 reproducible larger complete checks. These inherited records are distinct from the integration checks below; the unshared table below 100 million is not included in the independently verified evidence.

The integration candidate was internally hash-frozen before selecting fresh confirmation inputs; the commitment has no external timestamp. Development checked all 1,135 admissible primes below 20,000, with 3,405 complete method comparisons, 1,135 independent integer-binomial checks, 1,135 independently coded elliptic-prefix comparisons and 200 forced T3 checks. Three forest chunk counts agreed. Invalid domains and conductors, $p = 17$ and injected reconstruction misses were checked. The two relevant inherited test modules also passed, comprising 19 test methods.

Confirmation passed at 17 distinct primes, excluded from the recorded prior development, confirmation and benchmark ledgers, reaching $p = 7{,}000{,}000{,}009{,}553$. Every case compared complete native-factorial, nodal and automatic outputs and compared the Miller prefix with an independently coded elliptic provider. Applicable cases also used forced T3. The three approximately five-million inputs additionally received direct original weighted-recurrence, independent scaled-binomial-prefix and modular-binomial-product checks. At larger primes the original linear-time weighted recurrence was not run; agreement there uses independent prefix code and native-factorial boundary checks.

An exploratory forest survey produced all 39,264 admissible tuples below one million. The normalization and adjacent-sum residuals held throughout; neither $U_p(L)$ nor $U_p(m)$ vanished in that range. All 1,135 rows below 20,000 and 64 seeded rows above that bound received independent definition-level checks; the remaining rows

received internal identity checks. The nonexistence of simultaneous zeros follows separately from (48). This survey is newly executed evidence on a previously studied range, not disjoint confirmation. The integration package records the selection rules, hashes, exact outputs and raw measurements (Jordan, 2026a).

A subsequent normalization review validated the local accumulator against explicit rational Miller functions and checked the addition-law constants, including identity endpoints. For the uniform-order implementation accompanying Lemma 6 and Theorem 8, development passed 14,037 direct character-sum comparisons and confirmation passed 5,400, with no mismatches. Every value was compared with direct finite-field enumeration and the preceding corrected evaluator. Development included all squarefree cubics and quartics over $\mathbb{F}_5$, with nonzero leading coefficient and every nonbranch pole. An additional 47,712 nonbranch rational points on short cubics verified (U2) against explicit polynomial powering. The three executable source files were hashed before confirmation; nine confirmation primes were disjoint from this verifier's development prime set. These counts concern correctness, not new timing measurements, and overlapping exceptional-case counters are not independent samples. The proof, frozen code and JSON outputs accompany the revision (Jordan, 2026d).

Further questions

The core identity concerns admissible genus-one inputs with a fixed coefficient cutoff; the application section extends its interpretation to supplied simple-pole weights and elliptic-point sums. Projective arithmetic and a practical exact point-counting frontend could improve its implementation. Second-kind branch arguments, exceptional extension-field cases and other stopping families merit separate treatment. A higher-modulus congruence would require new arguments: the characteristic-p Cartier proof does not establish evaluation modulo p^2.

For the rounded-third family, a uniform polylogarithmic single-prime extractor for the scalar in (46), equivalently the Gauss factorial in (43), remains open. The completed integration shows that better boundary acquisition can improve the full weighted-sum evaluation without changing the Miller prefix. A precise next question is whether guaranteed reconstruction at larger conductors can reduce interval work without its setup cost erasing the gain. The fixed-conductor implementation and de Rham interpretation do not settle that question. Further implementation work includes machine-calibrated workload selection and platform validation. No general computational lower bound is claimed.

Limitations

The fixed-conductor rounded-third implementation retains square-root interval dependence, and certified acceptance does not guarantee successful lattice reconstruction. The benchmarks describe named implementations on one uncontrolled host with small samples; they establish neither universal crossovers nor superiority over all tools. The largest complete rounded-third check is approximately 43 bits. The regular quarter-point family has even trace and is not anomalous, so the trace-1 p-adic discrete-logarithm methods cannot be used as an independent cross-check without another comparison theorem. The native bridge was tested on Linux; Windows and future ABI compatibility remain unvalidated. Finite agreement and AI feedback are not specialist proof review. Our targeted primary-source comparison leaves publication priority for the combined scalar identity, residue-divisor formulation and weighted-sum adapters unresolved. The supplied simple-pole interface excludes active branch poles, higher-order poles and implicit denominator factorization from its logarithmic query bound. The uniform-order evaluator was checked for correctness but has not been assigned timings from the older implementation's benchmark.

AI assistance

Generative AI was used extensively in mathematical exploration, programming, computational checking and manuscript preparation under the author's direction. The reported computational checks arose within that research programme and the supplied AI feedback; they are not human peer review. The manuscript remains unrefereed by a human specialist.

Appendix on computational benchmarks

Original quarter-point pilot

Table 4 presents a quarter-point pilot for seven primes $p \equiv 1 \pmod 4$, with $n = (p-1)/4$. Both implementations compute B_n, a_n and $U_p(n)$, including per-input preparation. The comparator is a standalone C++ adapter to the unchanged SageMath 10.8 Harvey `hypellfrob` interval-product kernel, which uses BGS-based recurrence techniques (Bostan et al., 2007; Harvey, 2007; SageMath Developers, n.d.). This is a benchmark of that recurrence task, not of Harvey's full Kedlaya algorithm.

Table 4. Complete quarter-point evaluation in milliseconds: Python Cartier–Miller versus the C++ Harvey/BGS recurrence adapter, including per-input preparation

Prime p	Stopping index n	Cartier–Miller Python	Harvey BGS C++
97	24	0.05	0.01
1,009	252	0.06	0.05
10,009	2,502	0.07	0.10
100,049	25,012	0.09	0.54
1,000,033	250,008	0.10	1.78
10,000,121	2,500,030	0.11	4.44
100,000,037	25,000,009	0.29	16.19

The run used a shared hosted Linux environment reporting an AMD EPYC 9V74 processor, Python 3.12.14, GNU C++ 13.3.0 with `-O3`, and NTL 11.5.1 configured for one thread. Times are medians of 11 per-call timing-batch means, with alternating method order in warmed persistent processes. Batches targeted about 25 ms and were capped at 2,000 evaluations. Answers and per-call mathematical data were recomputed; library tables could remain warm. Validation, compilation, primality testing, process startup, communication and output writing were excluded. Host load and effective CPU frequency were not controlled.

All timed outputs agreed. Adapter validation additionally covered 10,809 stopping-point queries across 92 primes from 7 to 499, and 328 quarter-point comparisons for primes $p \equiv 1 \pmod 4$ from 13 to 4,999, with no mismatches. The timing run did not change these validation counts.

Cornacchia/Gauss preparation includes a nonresidue scan; this pilot does not implement Schoof setup or establish Corollary 4's worst-case bound. The direct characteristic-p proof did not change the evaluator. Small-input entries include call and loop overhead. Seven selected inputs, different implementation languages, an uncontrolled host and unmeasured memory limit general speed or scaling claims. Raw samples and reproduction commands accompany the manuscript.

Integrated rounded-third single-prime benchmarks

Tables 5–7 compare methods returning the identical complete tuple $(p, B_L, a_L, U_p(L), G)$. Each cell gives the median and observed minimum–maximum range of three samples with rotated method order on one host, using Python 3.12.14, python-flint 0.9.0 (FLINT 3.6.0), gmpy2 2.3.2 (GMP 6.3.0), Linux x86-64 and an AMD EPYC 9V74 processor. No concurrent number-theory job ran during timing; CPU frequency and scheduling were not controlled. These rounded-third measurements use a different protocol from the quarter-point pilot and must not be pooled with Table 4.

Each single-prime call rebuilds its Jacobi context and product trees and reruns Miller arithmetic; complete answers are not cached. Primality checks, registry and provider setup, conductor discovery, reconstruction certification, interval products, normalization, derived G and result collection are charged. Imports and prime selection are excluded, and reference comparisons occur after timing. A first call can load a prime-independent plan table or native-library handle. The native comparator calls FLINT's `n_factorial_fast_mod2_preinv` (FLINT developers, n.d.) for $m!$ and uses the same Miller prefix. The nodal comparator uses the same prefix with its factorial-ratio boundary.

Table 5. Complete rounded-third evaluation in milliseconds, shown as median [minimum, maximum] of three calls. Auto is the integrated automatic route; ℓ is its selected conductor.

Prime p	Auto route ℓ	Native + Miller	Nodal + Miller	Auto + Miller	Native / auto
70,000,283	T3 11	11.244 [11.013, 14.510]	9.716 [9.197, 11.064]	6.745 [6.105, 8.968]	1.67
70,000,481	T3 11	11.351 [11.004, 11.462]	9.547 [9.542, 11.833]	5.879 [5.386, 5.905]	1.93
70,000,613	T3 11	15.190 [14.775, 15.740]	12.261 [10.969, 12.432]	7.868 [6.523, 8.340]	1.93
7,000,000,013	T3 31	261.129 [254.152, 264.945]	220.053 [215.825, 251.297]	58.941 [56.682, 60.779]	4.43
7,000,000,277	nodal	256.361 [254.963, 259.401]	214.941 [208.206, 217.394]	214.746 [212.828, 215.143]	1.19
7,000,000,727	nodal	260.154 [255.525, 268.183]	214.452 [209.942, 215.550]	214.251 [212.405, 217.044]	1.21
7,000,000,769	nodal	263.522 [258.686, 265.291]	217.327 [212.077, 225.342]	225.696 [213.945, 226.674]	1.17
7,000,002,851	T3 23	259.000 [258.844, 273.559]	214.838 [213.861, 219.622]	63.885 [62.825, 65.143]	4.05
7,000,011,929	T3 13	266.407 [263.097, 283.127]	232.299 [218.851, 235.606]	77.799 [76.285, 85.781]	3.42
700,000,000,031	T3 71	5894.902 [5816.792, 6078.298]	4909.134 [4835.276, 4939.634]	1153.621 [1147.457, 1310.309]	5.11
700,000,001,633	T3 47	5833.161 [5831.019, 5857.249]	4760.048 [4749.239, 4880.084]	895.216 [868.507, 900.682]	6.52
700,000,004,549	T3 31	5751.378 [5718.851, 6103.010]	4784.071 [4749.360, 4909.331]	1014.534 [996.811, 1093.955]	5.67

All complete outputs agreed. At the nine inputs selecting T3, speedups over native factorial plus Miller ranged from 1.67 to 6.52. At $p = 700{,}000{,}001{,}633$, the complete median fell from 5.833161 s to 0.895216 s; the Miller prefix contributed approximately 0.48% of the new runtime. The no-small-factor inputs selected nodal fallback, so their gains over native are nodal gains, not T3 acceleration. Against direct nodal calls, automatic fallback has small selection overhead and timing variation.

Dense and sparse prime schedules

For Table 6 every method evaluates all admissible primes below the same exclusive bound. Sieve, arithmetic, collection and canonical tuple hashing are charged; disk I/O is excluded. Python and GMP forests use the same retained recurrence with different integer backends and acquire prefix and boundary jointly. Auto selects native throughout these small-prime ranges, so no dense-table gain is attributed to T3.

Table 6. Dense rounded-third tables, complete seconds as median [minimum, maximum] of three runs.

Exclusive bound	Tuples	Native + Miller	Auto + Miller	Python forest	GMP forest
2,000	152	0.120307 [0.120265, 0.122512]	0.122290 [0.120004, 0.126363]	0.001377 [0.001267, 0.001557]	0.001255 [0.001231, 0.001441]
20,000	1,135	1.273731 [1.272739, 1.344891]	1.336106 [1.278975, 1.344524]	0.015224 [0.013977, 0.016268]	0.011338 [0.010396, 0.017316]
100,000	4,805	6.677996 [6.650758, 6.870394]	6.652261 [6.646842, 6.750717]	0.157454 [0.155444, 0.161275]	0.057797 [0.056992, 0.057938]

At the 100,000 bound the GMP forest is approximately 115.5 times faster than independent native-factorial/Miller calls. This is the retained batch algorithm's amortization, not a speedup of the Miller pass. The forest need not be preferable for sparse schedules: it pays to construct recurrence products from index zero, rather than receiving the initial state near the first requested prime for free.

Table 7. One matched sparse schedule of sixteen primes from 1,000,211 to 2,501,171, complete runtime in seconds.

Method	Runtime in seconds
Native factorial + Miller	0.037191 [0.036596, 0.038203]
Automatic integration	0.037016 [0.036621, 0.039312]
Forced T3 $\ell = 5$	0.039155 [0.038894, 0.041857]
Python forest	8.171517 [7.951387, 8.624335]
GMP forest	0.689112 [0.679250, 0.817014]

The exact same sixteen-prime schedule and complete outputs were used throughout, with all output hashes agreeing. Independent native calls took 0.037191 s, versus 0.689112 s for the GMP forest, making the forest approximately 18.5 times slower here. Forced T3 at $\ell = 5$ did not improve this moderate-prime workload. Very large sparse-prime forest requests were outside the declared recurrence-cutoff cap of 2,000,000 and received no hypothetical timing.

The practical conclusion is workload-dependent: retain independent factorial and nodal alternatives, use certified Jacobi acquisition on suitable larger individual primes, and exploit a forest for dense tables. These experiments do not compare an optimized compiled forest, unmodified Tal reconstruction, the earlier project Jacobi backend, or complete PARI, SageMath and Magma systems. They establish improvements over the named matched comparators, not superiority over every available tool. The observed ranges are descriptive, not confidence intervals; in particular, modest fallback differences should not be interpreted as a robust universal speedup.

The reproducibility archive contains integrated.py, the original-definition oracles, the frozen selection rules, all 17 confirmation inputs, raw benchmark samples, the million-bound survey and preserved vendor proofs. README.md gives the commands for confirmation, benchmarks, survey and hash audit. The final file manifest and the earlier internal freeze have distinct roles: the freeze records the candidate before confirmation, whereas the manifest checks the delivered package. The standalone adapter integrates the existing boundary registry and complete-sum function; it does not constitute an updated graphical workbench release or a new many-prime dispatcher (Jordan, 2026a).

Matched character-sum workloads

This is a separate experiment with the identical residue $C_f(a)$, or identical multi-pole weighted sum, as output for every method in a comparison (Jordan, 2026c). Table 8 records medians for a cold mathematical instance: a fresh prime, curve and pole for each repetition, with trace preparation charged to the frozen Python Cartier–Miller implementation and no trace charged to the recurrence comparators. Imports, interpreter start, prime selection and PARI initialization were excluded. Outputs were checked outside the timed region.

Table 8. Cold single-pole character-sum medians in milliseconds. Within each cell, cubic / quartic. A dash means the method was not run at that class, not an extrapolated runtime. The native column is the best measured linear method in that class (C recurrence, C enumeration or FLINT polynomial power).

Bits	Frozen Cartier–Miller including trace	Harvey/BGS adapter	Best native linear
10	0.138 / 0.169	0.561 / 0.610	0.033 / 0.040
20	0.246 / 0.239	6.34 / 14.78	25.04 / 38.93
30	0.367 / 0.356	170.21 / 474.16	—
40	1.18 / 0.983	6,900 / 16,000	—
62	131.73 / 133.69	—	—

The uncontrolled Linux host reported an Intel Xeon at 2.10 GHz, two virtual CPUs and 7.8 GiB RAM. Software was CPython 3.13.16, GCC 13.3.0 with `-O2`, python-flint 0.9.0/FLINT 3.6.0, cypari2 2.2.4/PARI 2.17.2 without SEA data, and the Harvey `hypellfrob` 2.1.1 kernel through passagemath-ntl 10.8.13. Workers were single-threaded and methods ran sequentially in seeded random order. Pilots chose repetition counts targeting roughly 40 ms per block, capped at 200 repetitions and nine blocks; slower calls used fewer repetitions. The archive retains every block, repetition count, input class, validation status and budget decision.

These results support a sparse-query advantage for the named implementation on the tested inputs. They do not compare every possible implementation or prove universal crossover values. The square-root comparators had a declared 40-bit ceiling and were not run above it. PARI trace preparation used its available CM or baby-step/giant-step paths, so these are not SEA or Schoof timings. The separate trace-only class medians were approximately 122/117 ms at 62 bits; they came from other fresh instances and are not an exact decomposition of the cold medians. The uniform-order evaluator of Theorem 8 was developed after these measurements and has no new timing claim here. In the known-trace exceptional family, a direct Voloch–Semaev adapter was also faster than the frozen candidate by approximately 2.0–5.7 times, reflecting implementation and adapter overhead with the same asymptotic short-chain cost.

Reuse changes the comparison. At 16 bits a prepared full-polynomial route overtook repeated frozen scalar queries between 8 and 64 poles for cubics, and between 64 and 512 for quartics. At 20–23 bits the frozen evaluator was fastest at every tested query count up to 512. Dense-query tests on 12–18-bit curves instead favoured all-poles correlation or polynomial multipoint evaluation, by approximately 9–147 times over the tested Python scalar loops. For a fixed excluded set with only residue coefficients changing, all methods used the same cached dot product; at eight poles the measured coefficient stage was approximately 0.7–1.1 microseconds. Thus a speed claim must specify the output, setup and reuse regime.

For one weighted sum at 20 bits and up to 256 poles, the frozen evaluator was the fastest measured route; the tested polynomial method also paid 55–135 ms of preparation. Conversely, the tested Python second-kind branch implementation took roughly one second at 20 bits, where the trace-free BGS branch adapter took approximately 5–12 ms. No logarithmic branch-query claim follows from the nonbranch theorem.

The benchmark archive documents a host restart, reruns of only the missing jobs with the unchanged harness, and deterministic reconstruction of missing case metadata. Thirteen polynomial or multipoint jobs at 23 bits exceeded a declared 512 MiB working-memory target; that target was not enforced at runtime. They were flagged and none was the fastest in its cell. They supply no evidence of performance within that target. The source freeze covered the specified candidate files, not the entire timing and postprocessing harness. Independent review recomputed the displayed cold-query medians and ran bounded correctness comparisons; it did not rerun the full timing campaign. Raw data and these limitations remain part of the reproducibility record. These character-sum measurements must not be pooled with the earlier quarter-point or rounded-third protocols.

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